

Quantum and Classical Kernels under Distribution

Shift: Kernel Geometry Governs

Out-of-Distribution Robustness

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Abstract

Whether quantum kernels improve robustness under distribution shift is usually asked as a quantum-versus-classical question. We argue, and show empirically, that it is more productively asked as a question about kernel geometry. We present a controlled kernel-swap study in which the classifier, preprocessing, split construction, and family-internal model-selection logic are held fixed while only the kernel varies, instantiated on four security benchmarks spanning two modalities and three drift mechanisms (EMBER static malware; UNSW-NB15 and ToN-IoT network flows under synthetic covariate shift and natural regime drift), two classifier families (support vector classification and a Laplace-approximation Gaussian process classifier with calibrated uncertainty), and 72 principal settings with 15 repeated runs each; all quantum kernels are computed by exact statevector simulation, and no hardware claim is made. Against the customary linear and RBF baselines, fidelity-based quantum kernels improve out-of-distribution balanced accuracy nearly uniformly (69–71 of 72 settings; Wilcoxon $p < 10^{-9}$ on network data). Against a broader classical family, a median-heuristic Laplacian kernel neutralizes the advantage on EMBER—revealing that the operative property is not quantumness but geometry—yet when both families receive the same bandwidth-tuning freedom the quantum advantage re-emerges on

EMBER and remains dominant under natural network drift (54 of 54 settings under the probabilistic classifier, $p = 1.6 \times 10^{-10}$). A mechanistic analysis explains these outcomes: within each setting, out-of-distribution accuracy tracks the survival of centered kernel-target alignment under shift, a regularity that holds in 89–99% of 360 setting-seed units across all four datasets, with high effective rank of the training kernel as its structural precondition. The classical-favorable exceptions of the original comparison are precisely the settings where the selected quantum kernel collapses to low-rank, classical-like geometry. Beyond the specific quantum-versus-classical verdict, the study contributes a reusable, fully reproducible protocol for attributing robustness effects to kernel geometry, and a design principle—prefer kernels whose label-aligned geometry survives shift—that spans classical and quantum families alike.

Keywords: quantum machine learning; quantum kernels; distribution shift; out-of-distribution robustness; kernel methods; kernel-target alignment; Gaussian processes; malware detection; network intrusion detection

1 Introduction

Quantum kernel methods are one of the clearest and most actively studied directions in quantum machine learning. By embedding classical data into quantum states and defining similarity through state overlaps, they provide a principled way to study how quantum-induced feature geometry affects supervised learning while keeping the downstream classifier fixed [Havlíček et al. \(2019\)](#); [Schuld and Killoran \(2019\)](#); [Schuld \(2021\)](#). At the same time, the recent QML literature has made an equally important point: empirical conclusions about quantum models can depend strongly on the data regime, the encoding, and the evaluation protocol [Huang et al. \(2021\)](#); [Schuld et al. \(2021\)](#); [Gil-Fuster et al. \(2024\)](#); [Bowles et al. \(2024\)](#); [Schnabel and Roth \(2025\)](#). For that reason, claims about the usefulness of quantum kernels are most informative when they are established under controlled, reproducible, and carefully matched comparisons rather than through heterogeneous pipelines or loosely aligned baselines.

This need for control becomes even more important under distribution shift. In many applied settings, the relevant question is not only whether a model separates classes well under in-distribution (ID) conditions, but whether its behavior remains

reliable once the evaluation distribution changes. The broader distribution-shift and domain-generalization literature has repeatedly shown that apparent robustness gains can weaken, disappear, or become difficult to interpret when model-selection strategy, hyperparameter search, split construction, or evaluation standardization are not carefully controlled [Quiñonero-Candela et al. \(2008\)](#); [Zhou et al. \(2023\)](#); [Gulrajani and Lopez-Paz \(2021\)](#); [Koh et al. \(2021\)](#); [Yao et al. \(2022\)](#). A convincing robustness study must therefore satisfy two requirements at once: it must isolate the factor of interest tightly enough to support attribution, and it must evaluate that factor broadly enough to determine whether the observed behavior reflects a meaningful regime-level effect rather than a narrow configuration artifact.

These issues are especially acute in security applications. Drift is not incidental to the problem; it is part of the problem definition itself. Static PE malware classification has been widely studied through benchmarks such as EMBER [Anderson and Roth \(2018\)](#), while more recent work has highlighted the gap between laboratory evaluation and performance on drifted malware, as well as the difficulty of adapting with limited post-drift data [Barbero et al. \(2022\)](#); [Li et al. \(2025\)](#). The emergence of updated resources such as EMBER2024 further reflects an active push toward more realistic, broader, and more deployment-relevant malware evaluation [Joyce et al. \(2025\)](#). Network intrusion detection faces the same structural challenge: traffic distributions move as services, infrastructure, and attack campaigns evolve, and public benchmarks such as UNSW-NB15 and ToN-IoT expose exactly this kind of regime change [Moustafa and Slay \(2015\)](#); [Alsaedi et al. \(2020\)](#). Taken together, this literature suggests that a useful security-ML study should not merely report accuracy on a convenient split; it should examine how model choices behave when the evaluation explicitly targets shift, and it should do so across more than one data modality and drift mechanism.

Kernel methods are a natural setting in which to ask such questions cleanly. Because kernel-based support vector machines separate the choice of classifier from the

choice of similarity function, they offer a principled way to study how representational geometry affects ID and OOD behavior while keeping the learning machinery fixed [Cortes and Vapnik \(1995\)](#); [Schölkopf and Smola \(2002\)](#). This perspective becomes particularly interesting in quantum machine learning, where quantum kernels are defined through implicit feature maps into quantum Hilbert spaces and estimated via state overlaps or related fidelity-based similarities [Havlíček et al. \(2019\)](#); [Schuld and Killoran \(2019\)](#); [Schuld \(2021\)](#). From this viewpoint, quantum kernels are not a vague alternative model class, but a concrete family of kernel constructions whose empirical value can—and should—be tested under the same standards applied to classical kernels.

At the same time, the quantum-kernel literature has matured to the point where optimistic demonstrations and critical scrutiny must now be read together. Foundational work established the feature-space interpretation of quantum kernels and motivated them as expressive alternatives to classical similarities [Havlíček et al. \(2019\)](#); [Schuld and Killoran \(2019\)](#). Subsequent studies emphasized that performance in quantum machine learning can depend strongly on the data regime, the encoding, and the effective geometry induced by the model [Huang et al. \(2021\)](#); [Schuld et al. \(2021\)](#). More recent work has sharpened this message further: understanding generalization in quantum machine learning requires care beyond classical intuition, and benchmarking conclusions can be highly sensitive to protocol choices [Gil-Fuster et al. \(2024\)](#); [Bowles et al. \(2024\)](#); [Schnabel and Roth \(2025\)](#). This makes broad claims of superiority difficult to defend and reinforces the need for conservative, protocol-aware empirical studies.

The gap we address lies precisely at this intersection, and it has three parts. First, there remains limited evidence from controlled, variability-aware comparisons between classical and quantum kernel families in security-relevant settings where out-of-distribution robustness is part of the evaluation target: existing studies often differ

simultaneously in model class, preprocessing, search space, or data protocol, making it difficult to attribute observed gains to the kernel itself. Second, when such comparisons are attempted, the classical reference set is typically narrow—linear and RBF kernels under a single classifier—so an apparent quantum advantage may reflect the weakness of the baselines rather than a property of the quantum family; probabilistic classifiers with principled uncertainty quantification, and classical kernels with richer spectral structure, are usually absent. Third, and most fundamentally, empirical wins and losses are rarely accompanied by an account of *why* a kernel family should behave better or worse under shift, leaving results as observations in search of an explanation. For a question as specific as whether quantum kernels help under distribution shift, these confounds and omissions are not peripheral; they are decisive.

In this paper, we study that question through a strictly controlled *kernel-swap* protocol, instantiated broadly enough to close all three gaps at once. Within each experimental setting we hold the classifier, the preprocessing pipeline, the split definitions, and the family-internal model-selection logic fixed, and vary only the kernel. The protocol is applied to four security benchmarks spanning two data modalities and three drift mechanisms: EMBER static PE malware under two synthetic covariate-shift constructions [Anderson and Roth \(2018\)](#), and UNSW-NB15 and ToN-IoT network-flow scenarios under both an analogous synthetic construction and *natural* regime drift, in which the out-of-distribution pool consists of genuinely drifted traffic [Moustafa and Slay \(2015\)](#); [Alsaedi et al. \(2020\)](#). Two classifier families consume the same precomputed kernels: support vector classification, and a binary Laplace-approximation Gaussian process classifier [Rasmussen and Williams \(2006\)](#) that adds calibrated predictive probabilities—and hence principled uncertainty quantification under shift—without changing anything else in the pipeline. The classical family ranges from the customary linear and RBF baselines to polynomial, Laplacian, and Matérn kernels with train-only median-heuristic length scales; the quantum family

comprises fidelity kernels induced by commonly used feature-map families. Crucially, both families receive the same bandwidth-tuning freedom (a gamma sweep for RBF, an angle-scaling sweep for the quantum maps), so that neither side is compared at a fixed operating point the other is allowed to tune away from [Shaydulin and Wild \(2022\)](#); [Canatar et al. \(2023\)](#). The main evaluation comprises 72 principal settings (18 on EMBER, 54 on network flows), each repeated across 15 runs, with paired Wilcoxon signed-rank tests and Holm correction on all family-level comparisons.

Beyond measuring who wins, we ask why. For every kernel, dimension, and setting we compute geometry descriptors of the induced Gram matrices—the effective rank of the training kernel [Roy and Vetterli \(2007\)](#), centered kernel-target alignment on the ID and OOD splits [Cristianini et al. \(2001\)](#); [Cortes et al. \(2012\)](#), kernel concentration statistics, and the geometric difference between classical and quantum kernels [Huang et al. \(2021\)](#)—and test which of these quantities predict out-of-distribution behavior within and across settings. This mechanistic layer converts the benchmark from a scoreboard into an instrument for attribution.

The resulting picture is sharper than a quantum-versus-classical verdict, and more useful. Against linear and RBF baselines, fidelity-based quantum kernels improve OOD balanced accuracy nearly uniformly across all four datasets (69–71 of 72 settings under both classifiers). Against the extended classical family, the advantage on EMBER is neutralized by a median-heuristic Laplacian kernel—demonstrating that the operative property is geometric rather than quantum—but re-emerges under symmetric bandwidth tuning, and remains dominant on network flows, where under the probabilistic classifier the quantum family wins all 54 settings with gains exceeding run-to-run variability in the majority. In-distribution separability favors the quantum family in every EMBER setting regardless of the classical reference set. The mechanistic analysis explains all of these outcomes with one regularity: within a setting, OOD accuracy tracks the survival of centered kernel-target alignment under shift—positive in 89–99%

of 360 setting-seed units across all four datasets and both classifiers—with high effective rank of the training kernel as its structural precondition. The classical-favorable exceptions of the baseline comparison are exactly the settings where the selected quantum kernel collapses to a low-rank, classical-like geometry, and the Laplacian is exactly the classical kernel whose geometry most resembles the quantum maps.

Our contributions are fourfold. First, a controlled, reproducible kernel-swap protocol for robustness studies that extends across classifier families (including a probabilistic classifier with calibrated uncertainty), across a broad classical reference set with symmetric bandwidth tuning, and across datasets, modalities, and drift mechanisms. Second, a variability-aware regime map of quantum-versus-classical kernel behavior under shift on 72 settings with formal significance testing, including the honest negative result that a well-chosen classical kernel matches the quantum family on one modality at fixed bandwidth. Third, a mechanistic account—alignment survival under shift, enabled by high effective rank—that predicts within-setting OOD performance across every dataset we study, explains the structured exceptions, and yields the checkable prediction (which we confirm) that the classical kernels sharing this geometry are the competitive ones. Fourth, practical guidance and a fully public artifact: for OOD-sensitive kernel selection, prefer kernels whose label-aligned geometry survives shift; fidelity-based quantum maps exhibit this property natively, and on drifting network traffic they currently exploit it best.

2 Related Work

2.1 Distribution shift, domain generalization, and evaluation rigor

The broader literature on distribution shift has established that predictive performance can degrade substantially when training and test data are drawn from different

distributions, and that robust evaluation under shift requires more than reporting strong in-distribution performance [Quiñonero-Candela et al. \(2008\)](#); [Shimodaira \(2000\)](#); [Sugiyama et al. \(2007\)](#). Within machine learning, domain generalization (DG) has emerged as one of the main formal settings for studying out-of-distribution (OOD) performance without access to target-domain labels during training [Zhou et al. \(2023\)](#). This literature has also made clear that empirical conclusions about robustness are highly sensitive to evaluation design.

A particularly important development is the move toward benchmark-driven scrutiny of robustness claims: DomainBed showed that many reported domain-generalization improvements dissolve once datasets, architectures, hyperparameter budgets, and model-selection criteria are held consistent [Gulrajani and Lopez-Paz \(2021\)](#); WILDS demonstrated that standard training leaves large OOD gaps under naturally occurring real-world shifts [Koh et al. \(2021\)](#); and Wild-Time established temporal shift as a distinct, practically relevant regime that existing methods struggle to close [Yao et al. \(2022\)](#). These works matter here not because security detection is a standard DG benchmark, but because they establish a methodological principle we adopt wholesale: when the goal is to compare robustness under shift, fairness of protocol is part of the contribution. Our comparison therefore fixes the classifier family, preprocessing, and split definitions, so that the role of the kernel can be interpreted with minimal confounding.

2.2 Security detection under drift and evolving evaluation benchmarks

In malware detection, the relevance of distribution shift is especially acute. Malware ecosystems evolve continuously as attackers modify payloads, packers, obfuscation strategies, delivery mechanisms, and development toolchains. Static detection models

are therefore exposed to changing distributions in ways that are operationally consequential rather than merely academic. Public benchmarks have played an important role in making this problem more reproducible. EMBER remains one of the most widely used open datasets for static PE malware detection and helped standardize large-scale evaluation in this area [Anderson and Roth \(2018\)](#). More recent resources such as EMBER2024 push this effort further by broadening scope, increasing scale, and enabling more holistic evaluation across multiple malware-related tasks [Joyce et al. \(2025\)](#). Other datasets such as BODMAS and SOREL-20M likewise reflect the field’s effort to move toward larger and more realistic benchmark settings [Yang et al. \(2021\)](#); [Harang and Rudd \(2020\)](#).

At the same time, a growing body of work has shown that strong laboratory performance does not guarantee robustness to evolving malware. The Transcend line introduced conformal criteria to identify aging classifiers in deployment before their accuracy visibly degrades [Jordaney et al. \(2017\)](#), later refined by Barbero et al. [Barbero et al. \(2022\)](#); CADE detects and explains drifting samples without out-of-distribution labels [Yang et al. \(2021\)](#); and Li et al. studied adaptation to real drifted malware under limited post-drift supervision [Li et al. \(2025\)](#). On the methodological side, TESSER-ACT established that spatial and temporal experimental bias can manufacture illusory detector performance [Pendlebury et al. \(2019\)](#), and Arp et al. catalogued the recurring pitfalls of ML evaluations in computer security—sampling bias, label leakage, and inappropriate baselines among them [Arp et al. \(2022\)](#). Our protocol is, in large part, an application of those prescriptions to kernel comparison: fixed splits, leakage checks, temporal-style regime separation in the natural-drift construction, and baselines strengthened until they stop flattering the method under study. Taken together, these studies support a clear conclusion: in security detection, robustness under shift is not a niche desideratum but a primary evaluation target.

The same structural problem, with faster drift dynamics, defines network intrusion detection: benchmark corpora such as UNSW-NB15 and ToN-IoT capture traffic whose attack composition and background behavior change across collection regimes [Moustafa and Slay \(2015\)](#); [Alsaedi et al. \(2020\)](#), and adaptive NIDS pipelines are routinely retrained for exactly this reason. Using both modalities in one protocol lets us ask whether kernel-level robustness effects are properties of a dataset family or of the kernels.

Quantum machine learning has itself entered this application space. Quantum-enhanced classifiers and QSVMs have been applied to network intrusion detection since at least Gouveia and Correia [Gouveia and Correia \(2020\)](#), the area has been systematically reviewed [Nicesio et al. \(2023\)](#); [Nieto and Gil-Merino \(2026\)](#), and recent systems evaluate quantum kernels directly on the benchmarks we use, reporting quantum advantages on UNSW-NB15 and ToN-IoT [Abreu et al. \(2024, 2025\)](#); a critical evaluation on PCA-based intrusion detection found classical methods to remain competitive under realistic constraints [Bellante et al. \(2025\)](#). A common trait of the optimistic studies is decisive for the present paper: they compare against a small set of classical baselines on i.i.d. splits—precisely the two evaluation choices our results show to inflate apparent quantum advantages. Our study can be read as the missing stress test for this literature: under matched baselines, symmetric tuning, and explicit shift, part of the advantage survives and part of it does not, and the geometry explains which.

Our work differs from this line in emphasis. Rather than proposing a new drift-adaptation mechanism, drift detector, or continual-update strategy, we ask a narrower but complementary question: under a tightly controlled protocol, does the choice of kernel family itself materially affect OOD robustness? This question is well motivated by the security literature, yet comparatively underexplored in a way that isolates kernels from the many other moving parts of the pipeline.

2.3 Kernel methods, quantum kernels, and careful benchmarking in QML

Kernel methods provide a natural framework for studying robustness under controlled model comparisons because they separate the learning algorithm from the similarity function used to compare inputs. In classical machine learning, support vector machines and related kernel methods have long provided a principled route for controlling model capacity while varying the geometry of the induced feature space [Cortes and Vapnik \(1995\)](#); [Schölkopf and Smola \(2002\)](#). This makes kernels especially attractive when the scientific question concerns representation and similarity rather than wholesale changes in classifier architecture.

Quantum kernel methods arise from the observation that data can be embedded into quantum states and compared through overlaps in a quantum feature space. [Havlíček et al.](#) popularized this perspective in the context of supervised classification with quantum-enhanced feature spaces [Havlíček et al. \(2019\)](#), while [Schuld and Killoran](#) provided a broader theoretical account linking quantum models to kernel methods through feature Hilbert spaces [Schuld and Killoran \(2019\)](#). Later work sharpened this connection by arguing that supervised quantum machine learning models can often be interpreted directly as kernel methods, making the kernel viewpoint central rather than incidental [Schuld \(2021\)](#). Additional work on data encoding further showed that representational behavior depends critically on how classical inputs are mapped into quantum states, which is directly relevant when comparing quantum feature maps as alternative kernel constructions [Schuld et al. \(2021\)](#). Two spectral results frame what such kernels can and cannot do: [Kübler et al.](#) showed that the inductive bias of quantum kernels is governed by the spectrum of the induced integral operator, with flat spectra implying poor in-distribution sample efficiency [Kübler et al. \(2021\)](#), and [Thanasilp et al.](#) established that broad families of fidelity kernels concentrate exponentially toward trivial values as systems grow, erasing trainability [Thanasilp et al.](#)

(2024). Both results interact nontrivially with our findings and are taken up in the discussion.

However, the contemporary QML literature also makes clear that promising empirical behavior should not be read uncritically. Huang et al. showed that the role of data can be decisive in quantum machine learning, suggesting that observed performance gains are deeply regime-dependent rather than model-intrinsic Huang et al. (2021). Caro et al. studied generalization from few training examples, while Gil-Fuster et al. later argued that understanding generalization in QML requires rethinking familiar classical intuitions, since standard explanations may fail to account for observed behavior in quantum models Caro et al. (2022); Gil-Fuster et al. (2024). In parallel, broader perspective pieces have emphasized both the promise and the unresolved challenges of QML as a field Cerezo et al. (2022).

Most importantly for the present work, recent benchmarking studies have raised the standard for empirical claims in QML. Bowles et al. argued that experimental design choices can dominate conclusions and showed, in a broad simulated benchmark, that out-of-the-box classical models often remain strong baselines against quantum alternatives Bowles et al. (2024). Schnabel and Roth extended this benchmarking perspective specifically to quantum kernel methods, conducting a large-scale study across many design choices and showing that performance depends strongly on kernel type, encoding strategy, and hyperparameter configuration rather than on a simple classical-versus-quantum dichotomy Schnabel and Roth (2025). Related controlled evidence from adjacent QML settings points in the same methodological direction: recent work in hybrid quantum reinforcement learning shows that varying multiple pipeline components at once can obscure attribution, and highlights the importance of component-isolating, reproducible evaluation under fixed experimental conditions Lazaro et al. (2025). This line of work is especially relevant to our paper because it

supports a conservative framing: the value of quantum kernels, if any, must be demonstrated under controlled and transparent comparisons, and should be expected to be regime-dependent.

Three further strands supply the analytical tools of this study. First, work on kernel bandwidth in QML has shown that rescaling the data encoding is as consequential for fidelity kernels as the length scale is for classical kernels, and can determine whether quantum models generalize at all [Shaydulin and Wild \(2022\)](#); [Canatar et al. \(2023\)](#); [Eggering et al. \(2024\)](#); we operationalize this insight as a fairness requirement—symmetric bandwidth freedom for both families—rather than as a quantum-side tuning trick. Second, kernel-target alignment [Cristianini et al. \(2001\)](#); [Cortes et al. \(2012\)](#) and effective rank [Roy and Vetterli \(2007\)](#) are classical tools for characterizing Gram-matrix geometry, and the theory of spectral bias and task-model alignment shows that in-distribution generalization of kernel methods is governed precisely by the compatibility between the kernel’s spectrum and the target function [Bordelon et al. \(2020\)](#); [Canatar et al. \(2021\)](#); alignment has also served as a training objective for quantum embedding kernels [Hubregtsen et al. \(2022\)](#), a line our negative result on a-priori alignment-based selection speaks to directly. To our knowledge, the joint use of these descriptors as *predictors of OOD behavior under distribution shift*, across classical and quantum kernels alike, is new: we study the shift-facing counterpart of task-model alignment—whether the aligned spectrum survives once the distribution moves. Third, Gaussian process classification [Rasmussen and Williams \(2006\)](#) provides the standard route to calibrated uncertainty in kernel methods; including a Laplace-approximation GP classifier in the kernel-swap protocol lets us evaluate uncertainty quality under shift without changing anything else in the pipeline.

Finally, we note concurrent work. A 2026 preprint studies the robustness of quantum kernel methods under additive input corruption and proposes a spectral phase encoding to mitigate it [Herrero Gómez et al. \(2026\)](#). It shares our move away from

clean-data peak performance, but differs in all three defining aspects: it perturbs inputs with synthetic noise rather than evaluating under covariate or natural regime shift, it proposes an encoding rather than auditing families under a fixed protocol, and it does not use Gram-matrix geometry as the explanatory variable. Large-scale quantum-kernel benchmarking without varied train–test distributions [Bowles et al. \(2024\)](#); [Schnabel and Roth \(2025\)](#) remains the closest genre; the shift axis and the mechanism are the present paper’s additions to it.

Our study is positioned at the intersection of these literatures. From the distribution-shift literature, we adopt the principle that robustness claims must rest on evaluation protocols that control major confounds. From the security literature, we take the practical motivation that temporal and distributional change are intrinsic to realistic deployment, in malware and network intrusion detection alike. From the QML literature, we take both the motivation for studying quantum kernels and the caution that benchmarking quality is central to interpretation. Relative to prior work, our contribution is not to claim universal superiority for quantum kernels, nor to propose a new adaptation method for drifted security data, but to provide a controlled, variability-aware, multi-dataset comparison of classical and quantum kernel families in which out-of-distribution robustness is the primary evaluation target—and to accompany the comparison with a mechanistic account of why its outcomes are what they are.

3 Methods

3.1 Problem setting

We study binary security classification—malware versus benign software, and attack versus benign traffic—under distribution shift. Let D_{ID} denote the in-distribution data-generating process and D_{OOD} an out-of-distribution process induced by a shift mechanism, which may be synthetically constructed or naturally occurring. Given a

training set $T \sim D_{\text{ID}}$, we evaluate the learned classifier on both an ID test set S_{ID} and an OOD test set S_{OOD} . Our objective is not only to measure separability under the i.i.d. assumption, but to compare how kernel families behave when generalization is evaluated explicitly under shift.

The central methodological requirement of the study is attribution. Because the scientific question concerns the effect of the kernel, comparisons must avoid confounds from changes in classifier family, preprocessing, optimization protocol, or split design. We therefore adopt a strictly controlled *kernel-swap* protocol in which all components of the pipeline are held fixed except the kernel function itself. This design follows the same evaluation logic that motivates rigorous OOD benchmarking in the broader distribution-shift literature: when the target of inference is robustness, protocol control is not an implementation detail but part of the contribution.

3.2 Datasets and split construction

The protocol is instantiated on four benchmark scenarios spanning two data modalities. The first is EMBER, the standard public benchmark for static PE malware detection [Anderson and Roth \(2018\)](#), which is widely used, supports reproducible feature-based experimentation, and has motivated later benchmark extensions such as EMBER2024 [Joyce et al. \(2025\)](#). The remaining three are network-flow intrusion scenarios: two derived from UNSW-NB15 (with DoS and Reconnaissance, respectively, as the drifting attack category) [Moustafa and Slay \(2015\)](#) and one from ToN-IoT (Scanning) [Alsaedi et al. \(2020\)](#). In all cases our goal is not to benchmark absolute detection performance on a standard split, but to construct controlled ID/OOD regimes that allow kernel-family comparisons under distribution shift.

For each experimental setting, we adopt a fixed three-way split design:

$$T, \quad S_{\text{ID}}, \quad S_{\text{OOD}},$$

691 where T is used for training, S_{ID} measures in-distribution performance, and S_{OOD}
692 measures robustness under the induced shift. All models compared within a setting
693 use exactly the same split definitions. This point is essential: a classical and a quantum
694 model are never compared on different train/test partitions.
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698 ***EMBER shift constructions.***
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700 On EMBER we operationalize shift through two controlled OOD-construction vari-
701 ants, denoted $m1$ and $m2$, both built on the histogram + byte-entropy feature blocks
702 and both designed to preserve class balance across splits.
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705 ***Variant $m1$: sparsity-score extremes within class.***
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707 For each sample, we compute a sparsity-related score on a predefined feature block.
708 In the main configuration used in the experiments, this score corresponds to the num-
709 ber of non-zero entries within the selected block. The OOD set is formed by taking,
710 within each class separately, the low-score and high-score extremes, i.e., the two tails
711 of the within-class score distribution. The remaining samples form an ID pool from
712 which the training and ID test sets are obtained by stratified sampling. Intuitively,
713 this construction creates an OOD regime driven by atypical sparsity structure while
714 keeping the label distribution under control.
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718 ***Variant $m2$: distance-to-train extremes within class.***
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720 We first fix the final training set at an exact target size. Using only this training set,
721 we compute class-conditional centroids and assign each non-training sample a score
722 equal to its Euclidean distance to the centroid of its own class. To reduce noise and
723 avoid leakage, this distance can be computed in a TruncatedSVD-projected space
724 fitted on the training set only. The OOD set is then defined, again within each class
725 separately, by the low- and high-distance extremes from the non-training pool, with
726 exact target OOD size and proportional class allocation. The ID test set is drawn from
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the remainder. This produces a complementary shift regime in which OOD samples are unusual relative to the final training geometry rather than to a sparsity statistic alone.

Network-flow scenarios and shift constructions.

Each network scenario is defined by a *reference* pool (traffic from the pre-drift regime) and a *current* pool (traffic containing the drifted attack campaign), prepared with a numeric-only feature policy that removes identifiers, label-derived columns, and categorical protocol fields; UNSW-NB15 scenarios retain 39 flow features and ToN-IoT 89. We verified explicitly that no feature is label-derived: the maximum feature-label correlation across scenarios is 0.62 (attained by well-known legitimate features such as `sttl`), far from the near-perfect correlation that would signal leakage, and the verification script is part of the public artifact. On these pools we instantiate two shift mechanisms. The first, *m2-centroid*, is the direct analog of EMBER’s *m2*: within the class-balanced reference pool, a provisional training pool is fixed, class-conditional centroids are computed in a train-only MaxAbs+TruncatedSVD space, and the OOD pool consists of the within-class low- and high-distance tails. The second, *natural drift*, uses no synthetic construction at all: the training and ID pools are drawn from the reference regime and the OOD pool from the class-balanced current regime, so the evaluation measures robustness to genuinely drifted traffic rather than to constructed covariate extremes.

Together, these constructions are not intended as universal models of real-world drift. Rather, they provide reproducible shift mechanisms of three distinct kinds—atypical-sparsity extremes, train-geometry extremes, and natural regime drift—that are sufficiently different to support regime-level conclusions while preserving comparability across kernel families. In every setting, all models are compared on exactly the same T , S_{ID} , and S_{OOD} .

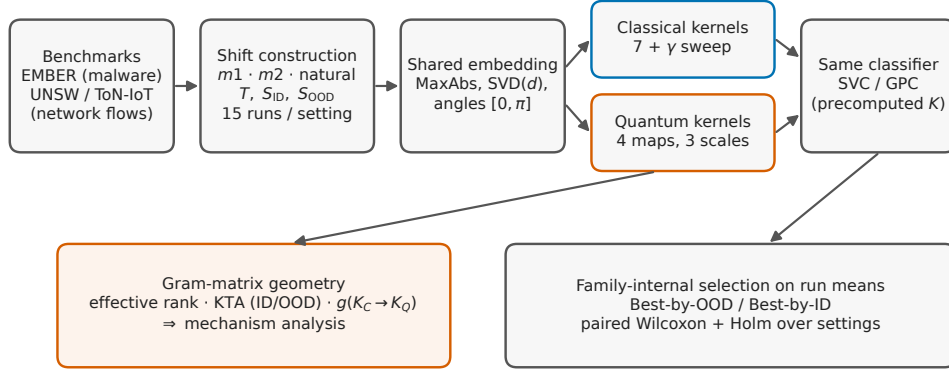


Fig. 1 The controlled kernel-swap protocol. Benchmark pools feed reproducible shift constructions; all kernels operate on a shared train-only embedding and are consumed by the same precomputed-kernel classifiers. Family-level conclusions are drawn by family-internal selection on run means with paired statistical testing, while the Gram matrices themselves feed the kernel-geometry analysis of Section 3.8.

3.3 Controlled kernel-swap protocol

Our core experimental principle is to isolate the kernel as the only variable of interest; Figure 1 summarizes the resulting pipeline. All compared models share the following components:

- **Classifier:** within each comparison, a single classifier consuming precomputed Gram matrices. Two classifier families are studied in parallel: Support Vector Classification with `kernel="precomputed"`, and a binary Gaussian process classifier under the Laplace approximation.
- **Preprocessing:** the same scaling and dimensionality-reduction pipeline for all classical and quantum models.
- **Split design:** identical T , S_{ID} , and S_{OOD} within each setting.
- **Selection logic:** the same family-internal model-selection principle applied consistently to classical and quantum candidates.

This yields a comparison between kernel families rather than between heterogeneous model pipelines. In particular, it prevents observed OOD differences from being

attributed to changes in classifier architecture, feature engineering, or split construction. The choice of precomputed-kernel classifiers is deliberate: it allows both classical and quantum similarities to be consumed by exactly the same decision rule, thereby making the comparison interpretable in kernel-method terms [Cortes and Vapnik \(1995\)](#); [Schölkopf and Smola \(2002\)](#).

The Gaussian process classifier implements the standard binary Laplace approximation with a logistic link (Algorithms 3.1 and 3.2 in [Rasmussen and Williams \(2006\)](#)), applied directly to the same unit-diagonal precomputed kernels as the SVC. Its role in the study is twofold. Methodologically, it tests whether the kernel-family conclusions are classifier-specific or transfer across learning machinery. Substantively, it produces calibrated predictive probabilities, which allows robustness under shift to be examined through the lens of uncertainty quantification—log-loss, Brier score, expected calibration error, and predictive entropy on ID versus OOD data—rather than through accuracy alone. Kernel-swap comparisons are always performed within a classifier, never across classifiers.

Although the classical and quantum candidate sets are not identical in cardinality, fairness in this study is defined at the protocol level: both families are evaluated under the same classifier, preprocessing, splits, dimensionality sweep, and family-internal selection criterion, so that the effect attributed to the kernel is not confounded by changes elsewhere in the pipeline.

3.4 Preprocessing and feature representation

All models use the same preprocessing pipeline. Let $x \in \mathbb{R}^p$ denote the original feature vector of the dataset at hand. We apply the following steps:

1. **MaxAbs scaling.** We first apply `MaxAbsScaler` to normalize feature magnitudes while preserving sparsity structure.

875 **2. Dimensionality reduction.** We project the scaled features using `TruncatedSVD`
876
877 to a reduced representation of dimension

878

879

880

$$d \in \{4, 6, 8, 10, 12\}.$$

881

882

883 **3. Angle mapping.** For the quantum pipeline, the reduced coordinates are linearly
884
885 mapped to an angle range $[0, \pi]$ before being passed to the quantum feature maps.
886

887 This shared preprocessing serves two purposes. First, it keeps the comparison fair
888
889 by ensuring that classical and quantum kernels operate on the same reduced repre-
890
891 sentation. Second, it makes representation size itself a controlled factor of the study,
892
893 enabling both aggregated setting-level comparisons and dimension-level analyses. The
894
895 dimensionality parameter d is therefore not a tuning artifact hidden in preprocessing,
896
897 but an explicit axis of the experimental grid.

898

899

3.5 Kernel families

900

901

Baseline classical kernels.

902

903

The customary reference set consists of two standard baselines:

904

905

906

$$K_{\text{lin}}(x, x') = x^\top x', \tag{1}$$

907

908

$$K_{\text{rbf}}(x, x') = \exp(-\gamma \|x - x'\|_2^2), \tag{2}$$

909

910

911 with the RBF kernel evaluated using the standard `scale` convention for γ and the
912
913 linear kernel cosine-normalized to unit diagonal. These are the baselines against which
914
915 quantum kernels are most commonly compared, and they define our *classical-original*
916
917 family.

918

919

920

Extended classical family.

Because an apparent quantum advantage over two baselines may say more about the baselines than about the quantum family, we additionally evaluate a broader *classical-extended* family: polynomial kernels of degree 2 and 3 (cosine-normalized), the Laplacian kernel $K(x, x') = \exp(-\|x - x'\|_1/\ell_1)$, and Matérn kernels with $\nu \in \{3/2, 5/2\}$, where all length scales ℓ are set by the median heuristic—the median pairwise distance on the training set, the standard choice in kernel two-sample testing [Gretton et al. \(2012\)](#)—so that every hyperparameter is determined from training data alone, without access to ID or OOD test information. These kernels were chosen deliberately to span a range of spectral behaviors: the Laplacian and low- ν Matérn kernels are less smooth than the RBF and induce Gram matrices with substantially heavier eigenvalue tails, which, as the mechanistic analysis will show, is precisely the property at stake under shift.

Bandwidth symmetry.

Kernel bandwidth is known to be a decisive hyperparameter for quantum kernels: rescaling the input encoding plays the same role for fidelity kernels that γ plays for the RBF kernel, and can qualitatively change generalization [Shaydulin and Wild \(2022\)](#); [Canatar et al. \(2023\)](#). A comparison in which the classical family may tune its bandwidth while the quantum family is pinned to a fixed encoding scale (or vice versa) is therefore not protocol-fair. In the bandwidth-symmetric portion of the study, both families receive the same tuning axis: the RBF kernel is evaluated at $\gamma = f \cdot \gamma_{\text{scale}}$ for $f \in \{0.1, 0.3, 1, 3, 10\}$, and every quantum feature map is evaluated at angle scalings $\{0.5, 1, 2\}$ of the canonical $[0, \pi]$ encoding. Family-internal selection then operates over the enlarged candidate sets exactly as before.

967 *Quantum kernels.*

968
969 Quantum kernels are defined through an implicit feature map $\phi_Q(\cdot)$ induced by a
970 parameterized quantum circuit, with similarity measured through fidelity-like overlaps
971 [Havlíček et al. \(2019\)](#); [Schuld and Killoran \(2019\)](#); [Schuld \(2021\)](#). For two inputs x
972 and x' , the corresponding kernel entry can be written abstractly as
973

$$K_Q(x, x') = |\langle \phi_Q(x), \phi_Q(x') \rangle|^2,$$

974
975
976
977
978 or equivalently through the overlap of states prepared by input-dependent quantum
979 feature maps.

980
981 In the experiments, we evaluate multiple commonly used fidelity-based feature-map
982 families chosen to represent distinct inductive biases: ZZ-type maps, PauliXZ-type
983 maps, and Z-map variants. Concretely, the candidate set includes two ZZ con-
984 figurations with different repetition depths, one PauliXZ configuration, and one
985 Z-map configuration. We emphasize that the study is not attempting to exhaust the
986 quantum-kernel design space; rather, it compares a representative set of commonly
987 used fidelity-based constructions under a tightly controlled protocol. This framing
988 is consistent with recent work arguing that empirical conclusions in QML are often
989 regime-dependent and highly sensitive to protocol choices [Schuld et al. \(2021\)](#); [Bowles](#)
990 [et al. \(2024\)](#); [Schnabel and Roth \(2025\)](#).
991

1000

1001

1002 **3.6 Model selection and repeated-run protocol**

1003

1004 A major source of ambiguity in robustness studies is the selection criterion. Strong ID
1005 performance does not necessarily imply strong OOD performance, and mixing the two
1006 objectives can make results difficult to interpret. We therefore use two complementary
1007 family-internal selection modes.
1008

1009

1010

1011

1012

Best-by-OD.

Within each family (classical or quantum), we select the configuration with the highest mean OOD balanced accuracy. We then compare the selected classical model against the selected quantum model. This is the robustness-oriented view and constitutes the primary comparison in the paper.

Best-by-ID.

Within each family, we instead select the configuration with the highest mean ID balanced accuracy and then compare the selected classical and quantum models. This is the separability-oriented view and is reported separately to avoid conflating ID accuracy with OOD robustness.

This separation is methodological, not cosmetic. It acts as a guardrail against over-interpreting ID gains as robustness gains and helps disentangle two distinct questions: which kernel family yields the strongest in-distribution class separation, and which yields the strongest out-of-distribution generalization.

Selection is only one source of ambiguity in robustness studies; a second is result stability. Each configuration is therefore repeated across 15 runs, generated by combining five q-split seeds with three model seeds under fixed setting definitions (shift variant, master seed, and size regime). Within a setting, both classical and quantum candidates are evaluated on identical split definitions and summarized by mean $\pm$ standard deviation over the repeated runs, so that every reported comparison is variability-aware while preserving strict comparability between kernel families.

Statistical testing.

Family-level comparisons are tested formally with paired two-sided Wilcoxon signed-rank tests over the per-setting deltas (the settings are the paired units), separately for each classifier and selection view, with Holm correction applied within each classifier across the reported comparisons. We complement these tests with the per-setting

effect-size proxy defined below, so that both the consistency of the sign pattern and the magnitude relative to run-to-run noise are visible.

3.7 Evaluation metrics

Our primary metric is balanced accuracy, reported separately on ID and OOD test sets:

$$\text{BAcc}_{\text{ID}}, \quad \text{BAcc}_{\text{OOD}}.$$

Balanced accuracy is preferred because it remains informative under moderate class imbalance and is straightforward to interpret in both settings.

The ID-to-OOD degradation

$$\text{drop} = \text{BAcc}_{\text{ID}} - \text{BAcc}_{\text{OOD}}$$

is recorded for every configuration and available in the artifact; because it conflates OOD behavior with the corresponding ID baseline, we treat OOD balanced accuracy as the primary inferential target and use drop only as a secondary descriptor.

To compare families after selection, we compute family deltas:

$$\Delta_{\text{OOD}} = \text{BAcc}_{\text{OOD}}^{(Q)} - \text{BAcc}_{\text{OOD}}^{(C)}, \tag{3}$$

$$\Delta_{\text{ID}} = \text{BAcc}_{\text{ID}}^{(Q)} - \text{BAcc}_{\text{ID}}^{(C)}. \tag{4}$$

Positive values indicate an advantage for the selected quantum model relative to the selected classical model.

Because average gains can be misleading when variability is large, we also report an effect-size-to-variability proxy:

$$\text{effect} = \frac{\mu_Q - \mu_C}{\sqrt{\sigma_Q^2 + \sigma_C^2}},$$

where μ_Q, μ_C and σ_Q, σ_C are the repeated-run means and standard deviations for the metric of interest. Values greater than 1 indicate that the observed mean gain exceeds the combined variability scale, which provides a simple and interpretable robustness check on whether an apparent gain is large relative to run-to-run noise.

Probabilistic metrics.

For the Gaussian process classifier we additionally report, on both ID and OOD splits, the negative log-likelihood (log-loss), the Brier score, the expected calibration error [Guo et al. \(2017\)](#) (ECE, 15 equal-width bins), and the mean predictive entropy. These quantities characterize how the quality and honesty of predictive uncertainty change under shift: a well-behaved probabilistic model should become *more* uncertain on drifted inputs (rising entropy) while its calibration degrades as little as possible (bounded ECE increase), the desideratum established by the benchmark of Ovadia et al. [Ovadia et al. \(2019\)](#).

3.8 Kernel-geometry descriptors

To move from measuring outcomes to explaining them, we compute geometry descriptors of the Gram matrices that every kernel induces in every setting and dimension. All descriptors are computed from the very kernel matrices consumed by the classifiers (unit diagonal by construction), so the analysis describes exactly the objects the models see.

1151 ***Effective rank.***

1152

1153 For the training Gram matrix K_{tr} with eigenvalues $\lambda_1 \geq \dots \geq \lambda_n \geq 0$ normalized

1154

1155 to $p_i = \lambda_i / \sum_j \lambda_j$, the effective rank is $\exp(-\sum_i p_i \log p_i)$ Roy and Vetterli (2007).

1156 It measures how many directions of feature space the kernel actually uses: a cosine-

1157

1158 normalized linear kernel on low-dimensional embeddings is near rank one, while fidelity

1159

1160 kernels can spread mass over many directions.

1161

1162 ***Centered kernel-target alignment.***

1163

1164 For a kernel block K on a labeled split with labels $y \in \{-1, +1\}^n$, the centered kernel-

1165

1166 target alignment (KTA) is $\langle K_c, yy^\top \rangle_F / (n \|K_c\|_F)$, where K_c is the doubly centered

1167

1168 kernel Cristianini et al. (2001); Cortes et al. (2012). We evaluate KTA on the training,

1169 ID, and OOD splits separately; the ID-to-OOD change in alignment measures whether

1170

1171 the label structure remains visible to the kernel’s geometry after shift—the quantity

1172

1173 we will call *alignment survival*.

1174

1175 ***Concentration and cross-split similarity.***

1176

1177 We record the mean and standard deviation of off-diagonal kernel entries within the

1178

1179 training block and of the cross blocks between each test split and the training set,

1180

1181 together with the ratio of mean OOD-to-train and ID-to-train similarities, which mea-

1182 sures how novel the OOD inputs look under each kernel’s similarity notion; these

1183

1184 cross-split means are the elementary statistics underlying kernel two-sample tests such

1185

1186 as MMD Gretton et al. (2012), used here descriptively rather than as hypothesis tests.

1187

1188 ***Geometric difference.***

1189

1190 Between each classical training kernel K_C and each quantum training kernel K_Q

1191

1192 we compute the regularized geometric difference $g(K_C \rightarrow K_Q) = \|\sqrt{K_Q}(K_C +$

1193

1194 $\lambda n I)^{-1} \sqrt{K_Q}\|_\infty^{1/2}$ of Huang et al. Huang et al. (2021), over a sweep of relative regu-

1195 larizers λ . Large values indicate that the quantum geometry can deviate substantially

1196

from what the classical kernel expresses; the quantity bounds the *potential* for a performance separation but not its sign.

Mechanistic analyses.

These descriptors feed two complementary analyses. First, a diagnostic analysis at the setting level asks which descriptors of the Best-by-OOD-selected kernels separate classical-favorable from quantum-favorable settings (Mann–Whitney AUC over settings) and correlate with the observed OOD delta. Second, a within-setting analysis asks whether geometry predicts performance across *all* kernel–dimension cells of a setting: for every setting, classifier, and q-split seed we compute the Spearman rank correlation between a descriptor (effective rank, OOD alignment) and OOD balanced accuracy over the 30–55 evaluated cells, and then examine the distribution of these correlations across settings and datasets. A descriptor that yields consistently positive within-setting correlations across datasets, drift mechanisms, and classifiers constitutes a dataset-general regularity rather than a benchmark artifact.

3.9 Experimental grid and reporting

The EMBER evaluation grid comprises 18 principal settings formed by the Cartesian product of:

- two shift variants ($m1, m2$),
- three master seeds,
- and three training-size regimes.

The three size regimes are:

$$(1000, 500, 500), \quad (2000, 1000, 1000), \quad (4000, 1800, 1800),$$

corresponding to $(n_{\text{train}}, n_{\text{ID}}, n_{\text{OOD}})$.

Table 1 Compact summary of the controlled experimental protocol. The repeated-run design is generated by combining five split seeds with three model seeds, yielding 15 runs per evaluated configuration under fixed setting definitions.

Component	Specification
Datasets	EMBER; UNSW-NB15 (DoS, Reconnaissance); ToN-IoT (Scanning)
Shift constructions	EMBER: $m1$, $m2$; network: m2-centroid, natural drift
Master seeds	42, 123, 999
Split seeds	42, 123, 999, 7, 2024
Model seeds	42, 123, 999
Runs per configuration	$5 \times 3 = 15$
Size regimes (n_{train} , n_{ID} , n_{OOD})	(1000, 500, 500), (2000, 1000, 1000), (4000, 1800, 1800)
Dimensions	$d \in \{4, 6, 8, 10, 12\}$
Classifiers	SVC (precomputed); Laplace-GP classifier (precomputed)
Shared preprocessing	MaxAbsScaler + TruncatedSVD + angle mapping
Classical-original family	Linear, RBF(scale)
Classical-extended family	+ poly(2,3), Laplacian, Matérn(3/2, 5/2), median heuristic
Quantum family	ZZ (1 rep), ZZ (2 reps), PauliXZ (1 rep), Z-map
Bandwidth sweep	RBF γ factors $\{0.1, 0.3, 1, 3, 10\}$; angle scales $\{0.5, 1, 2\}$
Selection views	Best-by-OOD, Best-by-ID (family-internal, on run means)
Statistical testing	Paired Wilcoxon over settings + Holm; effect proxy
Principal settings	EMBER $2 \times 3 \times 3 = 18$; network $3 \times 2 \times 3 \times 3 = 54$; total 72

The network-flow grid applies the same structure to the three scenarios (UNSW-DoS, UNSW-Reconnaissance, ToN-IoT-Scanning), each under two shift mechanisms (m2-centroid and natural drift), three master seeds, and the same three size regimes, yielding 54 principal settings. Together the study comprises 72 principal settings, each evaluated across the dimension sweep and repeated over 15 runs, for both classifiers. The bandwidth-symmetric sweep enlarges the kernel candidate sets on the full EMBER grid and on the natural-drift network settings.

For clarity and reproducibility, Table 1 summarizes the main experimental protocol used throughout the paper, including the shift constructions, controlled seed structure, size regimes, dimensionality sweep, kernel families, classifiers, and repeated-run design.

Within each setting, we evaluate both kernel families across the shared preprocessing grid and the dimension sweep $d \in \{4, 6, 8, 10, 12\}$. This produces both a setting-level comparison, used for the main Best-by-OOD and Best-by-ID analyses, and a broader dimension-level view used to characterize how representation size interacts with kernel-family behavior.

For reproducibility, split definitions are fixed within each setting, random seeds are controlled, and all key quantities are reported as mean $\pm$ standard deviation across repeated runs. Throughout the paper, we deliberately avoid claims of universal dominance or complexity-theoretic quantum advantage. The object of study is narrower and more empirical: under a fixed classifier and a fixed evaluation protocol, does changing only the kernel alter robustness under distribution shift in security-relevant tasks—and which property of the kernel is responsible?

4 Results

4.1 Overview

We report results under the controlled kernel-swap protocol described in Section 3. All comparisons share the same classifier, preprocessing pipeline, and ID/OOD split definitions within a setting; the only varying component is the kernel, and every number below is a mean over 15 repeated runs. The results are organized as an argument in four movements. First, on EMBER, the quantum family is compared against the customary linear+RBF baselines (Section 4.2). Second, the classical reference set is strengthened to the extended family, which changes the verdict on one modality (Section 4.3). Third, both families receive the same bandwidth-tuning freedom, which changes it back (Section 4.4). Fourth, the protocol is repeated on three network-flow scenarios including natural drift (Section 4.5). Section 4.6 then explains the entire pattern with a single geometric mechanism, and Section 4.7 examines predictive uncertainty under shift.

1335 4.2 EMBER versus the customary baselines: a near-uniform

1336

1337 quantum advantage

1338

1339 We begin with the comparison that dominates the literature: fidelity-based quantum

1340

1341 kernels against linear and RBF baselines. For each setting and classifier, family-internal

1342

1343 Best-by-OOD selection picks the configuration with the highest mean OOD balanced

1344

1345 accuracy on the run means, and the selected representatives are compared directly.

1346

1346 Table 2 (upper panel, first and third rows) reports the outcome.

1347

1348 The advantage is close to uniform. Under SVC the selected quantum kernel

1349

1349 improves OOD balanced accuracy in 17 of 18 settings (mean $\Delta_{\text{OOD}} = +0.037$, Holm-

1350

1351 adjusted Wilcoxon $p = 7.6 \times 10^{-5}$), with the gain exceeding the combined run-to-run

1352

1353 variability scale in 12 settings; under the GP classifier the corresponding figures

1354

1354 are 15 of 18, $+0.028$, $p = 2.1 \times 10^{-4}$, and 11 settings. The few classical-favorable

1355

1356 settings are marginal: all occur at the largest training size under the *m1* construc-

1357

1358 tion, with $|\Delta_{\text{OOD}}| \leq 0.010$ and effect sizes below one, i.e., within run-to-run noise.

1359

1359 The dimension-level view confirms that this is not a best-of-grid artifact: comparing

1360

1361 the best quantum against the best classical candidate separately at every setting-

1362

1362 dimension cell, the quantum family is ahead in 88 of 90 cells under SVC and 85 of 90

1363

1364 under the GP classifier (mean $\Delta_{\text{OOD}} \approx +0.05$ in both cases).

1365

1366 Under Best-by-ID selection (lower panel), the quantum family improves ID bal-

1367

1368 anced accuracy in all 18 settings under both classifiers, with mean gains of $+0.078$

1369

1369 (SVC) and $+0.065$ (GPC) and every setting exceeding the variability scale. We keep

1370

1371 the two views separate throughout: strong ID separability is not evidence of robustness,

1372

1372 and the following subsections show that the two properties respond very differently to

1373

1374 stronger classical baselines. Figure 2 previews the resulting arc: the per-setting OOD

1375

1376 deltas of this subsection (first column of each panel) collapse toward zero against

1377

1377 the extended family (second column), partially recover under bandwidth symmetry

1378

1379 (third), and re-emerge on network flows (fourth).

1380

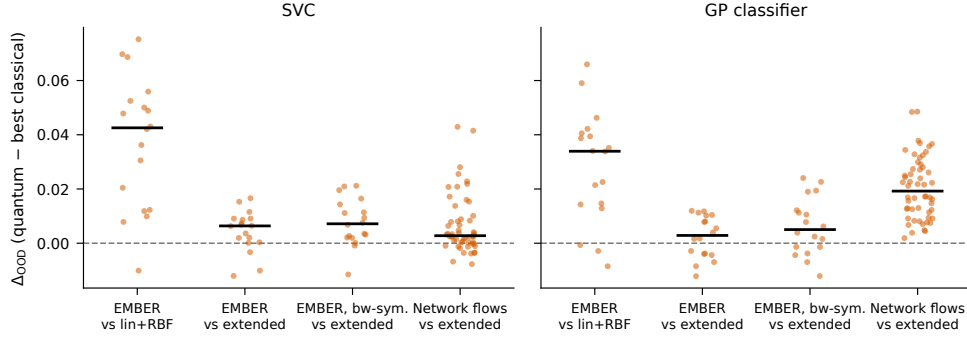


Fig. 2 The four movements of the study. Per-setting Best-by-OOO deltas (quantum minus best classical, mean over 15 runs), for SVC (left) and the GP classifier (right); black bars mark medians. Against linear+RBF the quantum advantage is large and nearly uniform; the extended classical family collapses it on EMBER; symmetric bandwidth tuning partially restores it; on network flows (54 settings, including natural drift) it persists against the extended family, uniformly so under the GP classifier.

Table 2 EMBER headline results (18 settings, 15 runs each). Within each classifier (GPC = Laplace Gaussian process classifier), family-internal selection compares the best quantum candidate against the best candidate of the stated classical reference set. p -values are two-sided paired Wilcoxon over settings, Holm-corrected within classifier; the last column counts settings whose quantum-favorable gain exceeds the combined run-to-run variability scale.

Selection	Classifier	Classical ref.	Q wins	Mean Δ	Median Δ	p (Holm)	Eff. > 1
Best-by-OOO	SVC	linear+RBF	17/18	+0.0374	+0.0426	7.6×10^{-5}	12
	SVC	extended	15/18	+0.0044	+0.0064	0.03	1
	GPC	linear+RBF	15/18	+0.0283	+0.0339	2.1×10^{-4}	11
	GPC	extended	11/18	+0.0023	+0.0029	0.26	0
Best-by-ID	SVC	linear+RBF	18/18	+0.0775	+0.0794	3.1×10^{-5}	18
	SVC	extended	18/18	+0.0149	+0.0152	3.1×10^{-5}	11
	GPC	linear+RBF	18/18	+0.0649	+0.0664	3.1×10^{-5}	18
	GPC	extended	18/18	+0.0192	+0.0166	3.1×10^{-5}	11

4.3 The extended classical family neutralizes the OOD advantage on EMBER

The comparison changes character when the classical family is strengthened. Against the extended set—which adds polynomial, Laplacian, and Matérn kernels with train-only median-heuristic scales—the quantum OOD advantage on EMBER collapses by an order of magnitude (Table 2, second and fourth rows): the mean delta falls from

+0.037 to +0.004 under SVC and from +0.028 to +0.002 under the GP classifier, the gain exceeds the variability scale in at most one setting, and under the GP classifier the comparison is no longer statistically significant (Holm-adjusted $p = 0.26$). The winner of this neutralization is overwhelmingly one kernel: the median-heuristic Laplacian is the Best-by-OOD selection of the extended classical family in 13 of 18 settings under SVC and 14 of 18 under the GP classifier, almost always at the largest embedding dimension.

Two aspects of this result deserve emphasis. First, it is an honest negative result for the quantum-advantage narrative as usually framed: the OOD gains reported against linear and RBF baselines—here and, we suspect, elsewhere in the literature—measure the weakness of those baselines as much as the strength of the quantum family. Second, the *ID* advantage does not collapse: even against the extended family, the quantum kernels improve ID balanced accuracy in 18 of 18 settings under both classifiers ($p = 3.1 \times 10^{-5}$), with the gain exceeding the variability scale in 11 settings. Whatever the quantum feature maps contribute to in-distribution separability, no classical kernel in our set replicates it; what the Laplacian replicates is specifically the out-of-distribution behavior. Section 4.6 shows that this is not a coincidence: the Laplacian is precisely the classical kernel whose Gram-matrix geometry most resembles that of the fidelity maps.

4.4 Bandwidth symmetry restores the advantage

The comparison of Section 4.3 pins every quantum feature map to the canonical $[0, \pi]$ encoding while granting the classical family kernels whose length scales adapt to the data. Bandwidth, however, is as decisive for fidelity kernels as it is for the RBF kernel (Shaydulin and Wild (2022); Canatar et al. (2023)). When both families receive the same tuning axis—RBF gamma factors $\{0.1, 0.3, 1, 3, 10\}$ against quantum angle scales $\{0.5, 1, 2\}$ —the picture inverts again (15 runs per setting, as in the main grid). Under

SVC, the quantum family beats the bandwidth-swept extended classical family in 16 of 18 EMBER settings (mean $\Delta_{\text{OOD}} = +0.008$, Holm-adjusted $p = 1.1 \times 10^{-3}$); under the GP classifier, in 12 of 18 ($+0.006$, $p = 0.03$); the gain exceeds the run-to-run variability scale in 4 settings under each classifier. The mechanism is visible in the selections: the angle-scaled variants account for 25 of the 36 Best-by-OOD quantum choices, confirming that the fixed-bandwidth comparison of Section 4.3 had the quantum family playing with one hand tied. On the network-flow scenarios under natural drift, the same symmetric sweep (one run per setting) leaves the quantum advantage intact and significant under both classifiers (24 of 27 settings, Holm-adjusted $p = 5.2 \times 10^{-5}$ for the GP classifier; 19 of 27, $p = 9.8 \times 10^{-4}$ for SVC).

The methodological moral generalizes beyond this study: fixed-encoding quantum kernels versus tuned classical kernels is not a protocol-fair comparison, and the bias runs *against* the quantum family. Once tuning freedom is symmetric, the extended classical family no longer neutralizes the quantum OOD advantage.

4.5 Network flows: the advantage survives the extended family and peaks under natural drift

The 54 network-flow settings test whether the EMBER conclusions are properties of one dataset family or of the kernels. They are neither, exactly: the network results are *stronger* for the quantum family than EMBER’s (Table 3). Against linear+RBF, the selected quantum kernel improves OOD balanced accuracy in 54 of 54 settings under both classifiers (Holm-adjusted $p = 6.5 \times 10^{-10}$), with the gain exceeding the variability scale in 35 (SVC) and 44 (GPC) settings. More importantly, against the *extended* classical family—the set that neutralized the advantage on EMBER—the quantum family remains ahead in 54 of 54 settings under the GP classifier (mean $\Delta_{\text{OOD}} = +0.020$, $p = 6.5 \times 10^{-10}$, effect > 1 in 33) and in 41 of 54 under SVC

Table 3 Network-flow headline results (54 settings across UNSW-NB15 DoS/Reconnaissance and ToN-IoT Scanning, two shift mechanisms, 15 runs each). Columns as in Table 2.

Selection	Classifier	Classical ref.	Q wins	Mean Δ	Median Δ	p (Holm)	Eff. > 1
Best-by-OOD	SVC	linear+RBF	54/54	+0.0256	+0.0264	6.5×10^{-10}	35
	SVC	extended	41/54	+0.0066	+0.0028	3.2×10^{-5}	7
	GPC	linear+RBF	54/54	+0.0344	+0.0338	6.5×10^{-10}	44
	GPC	extended	54/54	+0.0204	+0.0192	6.5×10^{-10}	33
Best-by-ID	SVC	linear+RBF	54/54	+0.0210	+0.0104	6.5×10^{-10}	29
	SVC	extended	39/54	+0.0036	+0.0028	1.5×10^{-4}	11
	GPC	linear+RBF	53/54	+0.0188	+0.0100	6.5×10^{-10}	24
	GPC	extended	40/54	+0.0057	+0.0044	7.9×10^{-7}	14

($p = 3.2 \times 10^{-5}$). The Laplacian is again the dominant classical challenger, and again it is not enough.

The breakdown by shift mechanism sharpens the picture. Under SVC, the quantum-versus-extended comparison is essentially tied on the synthetic centroid construction (15 of 27 settings) but nearly uniform under natural drift (26 of 27): the advantage is largest precisely on the drift mechanism that was not constructed by us and most resembles deployment conditions. Under the GP classifier the advantage is uniform on both mechanisms (27 of 27 each). Read jointly with Section 4.3, the regime map is: on EMBER’s synthetic covariate extremes a heavy-tailed classical kernel can match the quantum family at fixed bandwidth; on drifting network traffic, under a probabilistic classifier, nothing in our classical set does.

4.6 One geometric mechanism explains the pattern

The results so far pose a puzzle with three faces: why do quantum kernels beat linear and RBF baselines so uniformly, why does the Laplacian—alone among our classical kernels—catch them on EMBER, and why does natural network drift favor them again? The geometry descriptors of Section 3.8 answer all three at once.

Table 4 Within-setting Spearman correlation between kernel-geometry descriptors and OOD balanced accuracy, summarized per dataset and classifier over all setting-seed units (five q-split seeds per setting; 30–55 kernel–dimension cells per unit). “> 0” is the fraction of units with a positive correlation.

Dataset	Classifier	Units	Effective rank		KTA on OOD	
			median ρ	> 0	median ρ	> 0
EMBER	GP classifier	90	0.78	99%	0.68	89%
EMBER	SVC	90	0.83	98%	0.65	89%
ToN-IoT Scanning	GP classifier	90	0.84	100%	0.54	99%
ToN-IoT Scanning	SVC	90	0.75	100%	0.59	99%
UNSW-NB15 DoS	GP classifier	90	0.55	98%	0.51	96%
UNSW-NB15 DoS	SVC	90	0.55	90%	0.42	96%
UNSW-NB15 Recon.	GP classifier	90	0.26	98%	0.57	94%
UNSW-NB15 Recon.	SVC	90	0.19	83%	0.46	99%

A cross-dataset regularity.

Within every setting we correlate, across all evaluated kernel–dimension cells, each geometry descriptor with OOD balanced accuracy (Table 4 and Figure 3). The alignment-survival quantity—centered kernel-target alignment evaluated on the OOD split—predicts OOD accuracy within settings essentially everywhere: the within-setting Spearman correlation is positive in 89–99% of the 360 setting-seed units, with median ρ between 0.42 and 0.74, across all four datasets, both classifiers, and all three drift mechanisms. Effective rank of the training kernel—a train-only quantity—is an equally strong predictor on EMBER and ToN-IoT (median $\rho \approx 0.75$ –0.84), moderate on UNSW-DoS, and weak (though still predominantly positive) on UNSW-Reconnaissance. The natural reading is that high effective rank is the structural precondition—a kernel whose geometry uses only one or two directions has nothing with which to track label structure into the tails—while alignment survival is the proximate predictor of OOD performance.

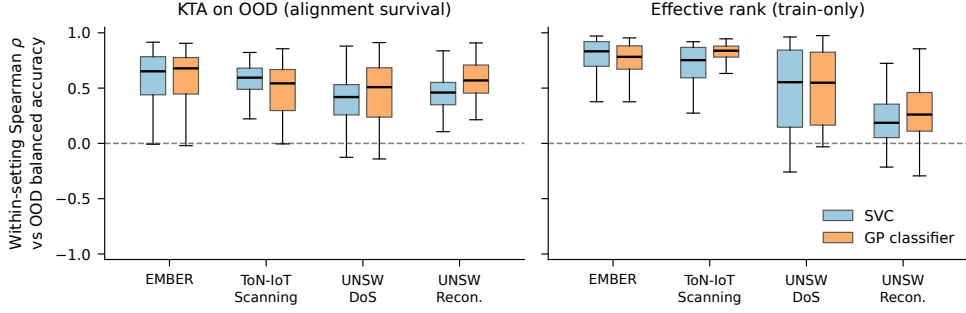


Fig. 3 The mechanism is a cross-dataset regularity. Distribution of within-setting Spearman correlations between each geometry descriptor and OOD balanced accuracy, over all setting-seed units, per dataset and classifier. Alignment survival (left) is positively correlated with OOD accuracy essentially everywhere; effective rank (right) is strong on EMBER and ToN-IoT and weaker, though still predominantly positive, on the UNSW scenarios.

A dose-response gradient within the classical family.

The mechanism makes a prediction that can be tested without any quantum kernel at all: classical kernels with higher effective rank should retain label alignment better under shift. Table 5 and Figure 4 confirm a clean monotone gradient across the seven classical kernels (18 EMBER settings $\times$ 5 dimensions each): as mean effective rank rises from 1.19 (linear) through 2.76 (RBF) and 6–8 (Matérn) to 17.7 (Laplacian), the mean ID-to-OOD change in alignment rises monotonically from -0.017 to $+0.019$, and the fraction of cells where alignment *improves* under shift doubles from 32% to 66%. This explains Section 4.3 precisely: the Laplacian is the classical kernel that shares the quantum maps’ geometric signature, and it is exactly the kernel that neutralizes their OOD advantage. Conversely, the fidelity maps’ near-uniform advantage over linear and RBF baselines is an advantage over the two lowest-rank geometries in the study.

What the geometry does not do.

Two negative results complete the mechanistic picture, and both are informative. First, the geometric difference of Huang et al. [Huang et al. \(2021\)](#)—the quantity a reader versed in the quantum-kernel literature would reach for first—does not predict which family wins a setting; at moderate regularization it even correlates *negatively* with the

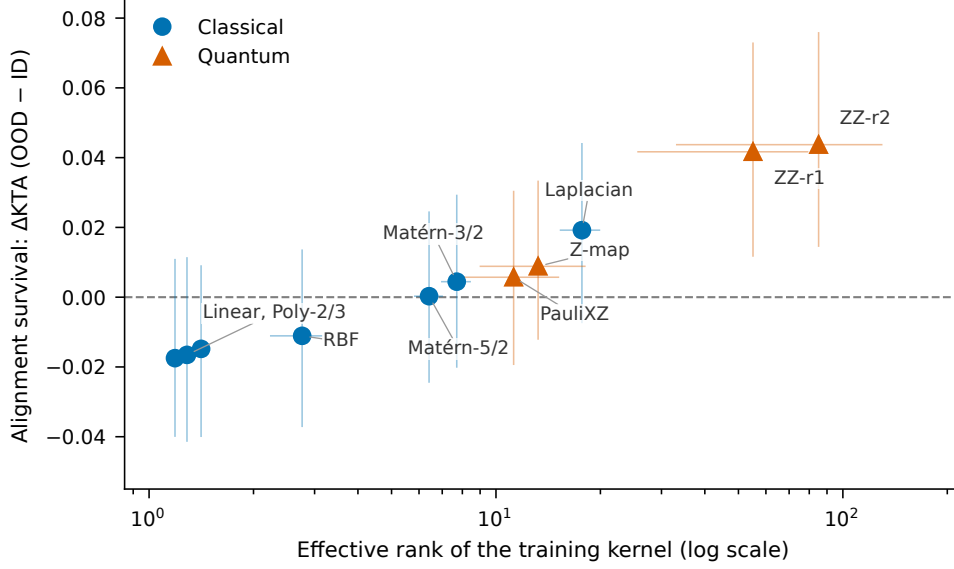


Fig. 4 Effective rank versus alignment survival across both kernel families. Markers are means over the 18 EMBER settings and 5 dimensions; bars span interquartile ranges. The relationship is monotone across both families: the classical kernels that retain alignment under shift are exactly the heavy-tailed ones, and the fidelity maps occupy the high-rank end of the same continuum.

Table 5 Dose-response gradient in the classical family (18 EMBER settings $\times$ 5 dimensions, q-split seed 42). Higher effective rank of the training Gram matrix is associated with better survival—indeed improvement—of kernel-target alignment from ID to OOD.

Kernel	Eff. rank	ΔKTA (OOD–ID)	Improves
Linear	1.19	−0.017	32%
Polynomial (2)	1.29	−0.017	32%
Polynomial (3)	1.41	−0.015	38%
RBF (scale)	2.76	−0.011	38%
Matérn 5/2	6.41	+0.000	46%
Matérn 3/2	7.71	+0.004	47%
Laplacian	17.69	+0.019	66%

observed OOD delta. This is consistent with its definition: g bounds the potential for a separation between the families, not its sign, so a quantum kernel can differ greatly from the classical reference and be worse. Second, we found no reliable fully a-priori selection rule: choosing the quantum configuration by training-set alignment alone

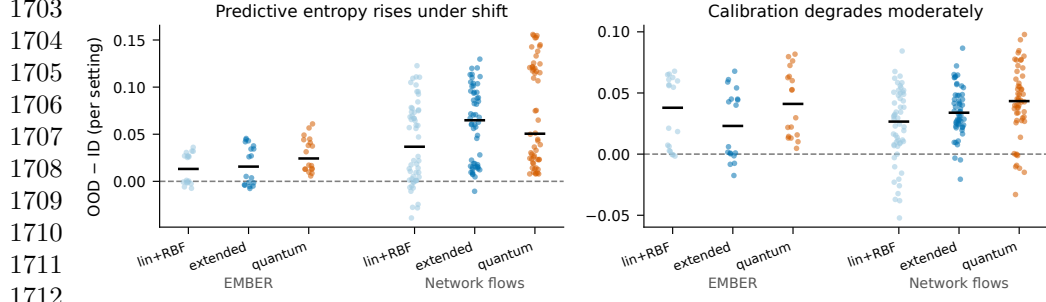


Fig. 5 Predictive uncertainty under shift. Per-setting ID-to-OD change in mean predictive entropy (left) and expected calibration error (right) for the Best-by-OD configuration of each family under the GP classifier; black bars mark medians.

does not identify the settings where the family will win. The geometry explains and diagnoses; it does not yet provide a deployment-time oracle, and we flag this honestly as an open problem.

4.7 Predictive uncertainty under shift

The GP classifier’s calibrated probabilities add a complementary robustness lens (Figure 5). Averaged over Best-by-OD selections, predictive entropy rises from ID to OOD for every family and dataset group—the qualitative behavior one wants from a probabilistic detector facing drift—and rises most for the quantum kernels (e.g., +0.029 versus +0.014–+0.018 for the classical families on EMBER; +0.072 versus +0.041–+0.061 on network flows). Calibration degrades moderately for all families (ECE increases of +0.02–+0.05), with the quantum family’s ECE increase slightly larger than the classical families’. In other words, the quantum kernels’ OOD accuracy advantage does not come with hidden overconfidence—their uncertainty grows under shift at least as fast as the classical kernels’—but neither do they calibrate better; the advantage is in the decision geometry, not in the probability calibration.

5 Discussion

5.1 What is, and is not, a quantum advantage here

The study’s central interpretive move is to replace the question “do quantum kernels beat classical kernels under shift?” with “which property of a kernel governs its behavior under shift, and who has it?”. Read through that lens, the results decompose cleanly. The near-uniform advantage over linear and RBF baselines is real, replicable across four datasets and two classifiers, and statistically unambiguous—but it is an advantage of *high-effective-rank, alignment-preserving geometry* over the two lowest-rank geometries in the study, a property the fidelity maps possess natively rather than exclusively. The Laplacian result makes this precise: one classical kernel with the same geometric signature closes the OOD gap on EMBER, so the property, not the physics, is doing the work there. What remains after every control we could devise—extended baselines, symmetric bandwidth tuning, a probabilistic classifier, natural drift—is a genuine residual: the ID-separability advantage survives everything on EMBER, and the OOD advantage survives everything on drifting network traffic. Those two residuals are, on present evidence, the defensible quantum contribution.

From the perspective of the distribution-shift literature, this reinforces a general lesson: conclusions about robustness are only as credible as the protocol used to obtain them [Quiñonero-Candela et al. \(2008\)](#); [Gulrajani and Lopez-Paz \(2021\)](#); [Koh et al. \(2021\)](#); [Yao et al. \(2022\)](#). Our results add a QML-specific corollary: the composition of the classical reference set and the symmetry of tuning freedom are not implementation details but the difference between a significant advantage, a null result, and a significant advantage again (Sections 4.2–4.4). Any single one of those three verdicts, reported alone, would have been misleading.

1795 5.2 The ID–OOD dissociation is informative

1796
1797 The contrast between the two selection views is sharper in this study than in its pre-
1798 decessors, and it now has an interpretation. The quantum ID-separability advantage
1800 is universal (18 of 18 EMBER settings against every classical reference, with effects
1801 exceeding run-to-run noise in most) and is not replicated by any classical kernel we
1802 evaluate, including the Laplacian. The OOD advantage, by contrast, is contingent: it
1803 depends on the reference family, the bandwidth protocol, and the drift mechanism.
1804 This dissociation is exactly what the mechanism predicts. In-distribution separability
1805 rewards expressive geometry per se; robustness rewards geometry whose label align-
1806 ment *survives* the shift, which is a joint property of the kernel and the shift mechanism.
1807 A kernel family can therefore dominate the first criterion while merely competing on
1808 the second—and mistaking one for the other is precisely the error the two-view pro-
1809 tocol is designed to prevent [Gulrajani and Lopez-Paz \(2021\)](#). The dissociation also
1810 speaks to the “accuracy on the line” literature, which documents a strong empiri-
1811 cal coupling between ID and OOD accuracy across models and the settings where it
1812 breaks [Miller et al. \(2021\)](#): at the level of kernel families under our shift constructions,
1813 the coupling breaks systematically, and the geometry identifies the axis along which
1814 it does.

1825 1826 5.3 The mechanism in context

1827
1828 The alignment-survival regularity connects several strands of literature that have so far
1829 run in parallel. The bandwidth results extend the findings of Shaydulin and Wild [Shay-](#)
1830 [dulin and Wild \(2022\)](#) and Canatar et al. [Canatar et al. \(2023\)](#) from in-distribution
1831 generalization to robustness under shift, and give them a comparative twist: band-
1832 width is not merely important for quantum kernels, it is important *symmetrically*,
1833 and asymmetric tuning protocols systematically bias quantum-versus-classical com-
1834 parisons. The failure of the geometric difference [Huang et al. \(2021\)](#) to predict OOD
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outcomes sharpens its interpretation rather than contradicting it: g was derived as a bound on the potential for in-distribution separations, and our results show empirically that potential-for-difference and benefit-under-shift are distinct quantities—large g is compatible with large losses. Finally, the dose-response gradient within the classical family (Table 5) grounds the mechanism independently of anything quantum: kernels with heavier spectral tails hold their label alignment into the tails of the data distribution, a statement about kernels generally, for which the quantum maps are simply the most extreme instances in our study.

The high-rank finding must also be reconciled with two spectral results that superficially point the other way. Kübler et al. argue that quantum kernels generalize in-distribution only when the relevant spectral mass is concentrated in few components Kübler et al. (2021), and Thanasilp et al. show that fidelity kernels concentrate exponentially—toward effectively rank-one, uninformative Gram matrices—as qubit counts grow Thanasilp et al. (2024). There is no contradiction, but there is a genuine tension that our results locate precisely. Both results concern the asymptotic or i.i.d. regime; our effective ranks (roughly 10–150 on training sets of 10^3 – 4×10^3 samples at 4–12 qubits) sit far below the concentration regime, and the studied maps demonstrably retain in-distribution power (the 18-of-18 ID advantage). Within that benign band, the quantity that governs *robustness* is how much of the spectrum survives with its label alignment intact under shift—and there, more rank helps. The bandwidth sweep makes the trade-off operational: angle scaling moves kernels along the rank axis, and the selected scales are those that raise rank without approaching the concentration pathology. Alignment-preserving rank is therefore best understood as bounded from both sides—enough spectral richness to track label structure into the tails, not so much that the kernel concentrates into noise.

We emphasize what the mechanism does not yet deliver. It is explanatory and diagnostic—given the Gram matrices of a setting, it accounts for which kernels will

1887 generalize under the shift—but our attempts to convert it into an a-priori selection
1888 rule from training data alone failed (Section 4.6). Alignment on OOD inputs requires
1889 OOD labels; effective rank is train-only but is a precondition, not a guarantee. Turning
1890 alignment survival into a deployable model-selection signal, perhaps via unlabeled
1891 OOD data or distributionally robust surrogates, is in our view the most interesting
1892 open problem this study raises.

1896

1898 5.4 Implications for security machine learning practice

1899

1900 From a practical perspective, the paper suggests three takeaways for security ML.
1901 The first is substantive: for OOD-sensitive kernel selection, prefer kernels whose label-
1902 aligned geometry survives shift. Concretely, if quantum kernels are unavailable or too
1903 costly, a median-heuristic Laplacian kernel is a far stronger robustness baseline than
1904 the RBF kernel that pipelines default to—on our EMBER settings it recovers most of
1905 what the fidelity maps offer. Where quantum kernels are on the table, they warrant
1906 serious consideration for drifting-traffic workloads, where their advantage was largest
1907 and most uniform in our study.

1912

1913 The second is methodological. Independently of the preferred kernel family, the
1914 protocol used here provides a reusable blueprint for robust evaluation in security
1915 settings:

1917

- 1918 1. keep the classifier and preprocessing fixed, and swap only the kernel;
- 1919 2. define explicit OOD splits tied to interpretable shift mechanisms, including at least
1920 one natural drift mechanism where the data permit;
- 1921 3. give every family the same hyperparameter freedom, bandwidth included;
- 1922 4. repeat runs, report variability, and test family-level claims formally;
- 1923 5. and separate model-selection criteria according to the actual objective (robustness
1924 vs. separability).

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Security ML frequently suffers from comparisons in which model class, tuning effort, and evaluation conditions change simultaneously; the three-verdict sequence of Sections 4.2–4.4 is a concrete demonstration of how much the conclusion depends on getting this right.

The third concerns diagnostics: the geometry descriptors are cheap to compute from Gram matrices that the pipeline already builds, and the alignment-survival analysis can be run post hoc on any deployed kernel system with a labeled drift sample. Practitioners should nonetheless resist reading these results as a blanket recommendation to replace classical kernels; the responsible recommendation is to test the candidate families under the expected shift regime, using exactly the kind of protocol described here.

5.5 Limitations and threats to validity

Several limitations define the scope of our conclusions.

Shift operationalization.

Our OOD splits cover three reproducible mechanisms—within-class sparsity extremes, train-geometry extremes, and natural regime drift on network traffic—but not the full space of real-world drift processes. Explicit temporal splits on timestamped malware, family-based drift, and adversarially induced shift remain outside the study’s scope, and the alignment-survival regularity should be re-tested under them rather than assumed.

Kernel and encoding coverage.

The classical family now spans seven kernels plus a bandwidth sweep, and both families receive symmetric tuning freedom, but neither candidate set is exhaustive: the quantum side is restricted to four commonly used fidelity feature maps at low qubit counts, and trained or task-adapted encodings, projected kernels, and deeper maps are

not evaluated. The mechanism gives a concrete prediction for such extensions—what should matter is their Gram-matrix geometry, not their provenance—which future work can falsify. At substantially larger qubit counts, exponential kernel concentration [Thanasilp et al. \(2024\)](#) would erode the alignment-preserving rank the mechanism rewards, so our conclusions should not be extrapolated beyond the low-qubit regime studied here.

Selection over candidate sets.

Best-by-OOD and Best-by-ID are selection procedures over finite candidate sets, applied identically to both families and evaluated on run means with formal paired testing. Residual sensitivity to the candidate space and search granularity nonetheless remains, and nested or cross-validated selection protocols would strengthen the analysis further.

Simulation, not hardware.

All quantum kernels are computed by exact statevector simulation. The study therefore characterizes the geometry of fidelity kernels, not the behavior of noisy devices: finite sampling of kernel entries, hardware noise, and error mitigation would all perturb the Gram matrices [Wang et al. \(2021\)](#), and exponential concentration implies that with polynomially many shots the very spectral richness our mechanism rewards becomes expensive to resolve at larger scales [Thanasilp et al. \(2024\)](#); the effect of either on alignment survival is unknown and worth measuring. We make no claim about hardware feasibility, resource advantage, or quantum speedup—the only rigorously established quantum-classical learning separation concerns a specifically constructed problem, not natural data [Liu et al. \(2021\)](#). At the dimensions studied, the fidelity kernels are also classically simulable [Schreiber et al. \(2023\)](#), which is precisely what makes them available as practical kernel constructions today.

Dataset scope and feature caveats.

Four benchmark scenarios across two modalities is a substantial widening relative to single-dataset studies, but security is broader still: dynamic malware analysis, encrypted traffic, and non-security modalities remain untested. Within the network scenarios, the feature sets inherited from the public benchmarks retain port numbers and protocol indicators, which are legitimate but known shortcut-prone features in NIDS; because every kernel receives identical features, this affects absolute accuracies, not the family comparisons. Finally, the mechanism’s effective-rank component is itself regime-dependent (strong on EMBER and ToN-IoT, weak on UNSW-Reconnaissance), a gradation we report rather than smooth over.

5.6 Conservative interpretation

In light of these limitations and the broader QML benchmarking literature, the appropriate interpretation of the paper is deliberately conservative. We do not claim universal superiority, complexity-theoretic quantum advantage, or hardware relevance. We claim, with formal statistical support, three narrower things: that fidelity-based quantum kernels beat the customary baselines under shift essentially wherever we looked; that this fact is explained by a geometric property a well-chosen classical kernel can partially share; and that under symmetric protocols the quantum family retains a significant OOD advantage on drifting network traffic and a universal ID-separability advantage on EMBER. This regime-aware view is consistent with recent calls for careful benchmarking in QML [Bowles et al. \(2024\)](#); [Schnabel and Roth \(2025\)](#) and, we believe, is scientifically stronger than either an unqualified advantage claim or an unqualified debunking: it says what the advantage is made of, where it lives, and how to check for it.

2071 6 Conclusion

2072

2073 This paper asked why kernel families behave differently under distribution shift,
2074 using the quantum-versus-classical comparison as both motivation and stress test. A
2075 controlled kernel-swap protocol—classifier, preprocessing, splits, selection logic, and
2076 tuning freedom held fixed while only the kernel varies—was instantiated on four secu-
2077 rity benchmarks spanning two modalities and three drift mechanisms, two classifier
2078 families including a probabilistic one, and 72 principal settings with 15 repeated runs
2079 and formal paired testing.

2085 The empirical arc has four movements. Fidelity-based quantum kernels beat the
2086 customary linear and RBF baselines under shift nearly everywhere. A stronger classical
2087 family neutralizes that advantage on EMBER, revealing that the operative property
2088 is geometric rather than quantum. Symmetric bandwidth tuning restores it, revealing
2089 that the neutralization itself depended on an asymmetric protocol. And on drifting
2090 network traffic the advantage is uniform under the probabilistic classifier, while the
2091 in-distribution separability advantage survives every control on EMBER. A single
2092 mechanism accounts for the whole pattern: kernels whose training geometry has high
2093 effective rank retain their kernel-target alignment under shift, and alignment survival
2094 predicts out-of-distribution accuracy within settings in 89–99% of 360 setting-seed
2095 units across all four datasets. The mechanism also explains the exceptions—the com-
2096 petitive classical kernels are exactly those sharing the geometric signature—and makes
2097 falsifiable predictions for kernel constructions we did not test.

2106 The contribution is therefore threefold. Empirically, the study provides the
2107 broadest controlled, variability-aware regime map of quantum-versus-classical ker-
2108 nel behavior under shift that we are aware of, including the honest negative results
2109 (the Laplacian neutralization, the non-predictiveness of the geometric difference, the
2110 absence of an a-priori selection rule). Mechanistically, it converts an observation into
2111 an explanation with cross-dataset support. Methodologically, it leaves behind a fully
2112

reproducible protocol—and a cautionary demonstration that the quantum-versus-classical verdict flips twice as the protocol is made progressively fairer, which we offer as a concrete standard for future QML benchmarking.

The most important open problems are two. Practically, turning alignment survival into a deployment-time selection signal without OOD labels would convert the diagnosis into a tool; the security community’s label-free drift detectors [Yang et al. \(2021\)](#) suggest this is not hopeless. Theoretically, the dose-response relationship between spectral tails and alignment survival calls for a formal account, and the ingredients exist: task-model alignment theory explains in-distribution kernel generalization through spectrum–target compatibility [Bordelon et al. \(2020\)](#); [Canatar et al. \(2021\)](#), and recent minimax analyses characterize RKHS regression under covariate shift with bounded density ratios [Ma et al. \(2023\)](#). Extending that line to the survival of aligned spectral mass under the shift models studied here—asking under which conditions high effective rank provably protects kernel-target alignment—is a concrete research program rather than a gap. Answering either question would benefit classical and quantum kernel methods alike—which is, in the end, the point: the productive question is not whether one family dominates, but which similarity geometry aligns best with the shift regime one actually faces.

Appendix A Additional Results

This appendix reports the full setting-level Best-by-OOD comparisons between the selected quantum kernel and the best kernel of the *extended* classical family (the strongest comparison in the study), for both classifiers and all 72 principal settings. Every value is a mean $\pm$ standard deviation over 15 repeated runs; Effect denotes $\Delta_{\text{OOD}}/\sqrt{\sigma_C^2 + \sigma_Q^2}$. Size labels denote $(n_{\text{train}}, n_{\text{ID}}, n_{\text{OOD}})$: S = (1k, 500, 500), M = (2k, 1k, 1k), L = (4k, 1.8k, 1.8k). Shift labels for the network scenarios abbreviate scenario/mechanism: UD = UNSW-NB15 DoS, UR = UNSW-NB15

2163 Reconnaissance, $TS = \text{ToN-IoT Scanning}$; $/m2$ is the centroid construction and
2164 $/nat$ is natural drift. Quantum kernel names carry the angle scale after $@$ where
2165 it differs from 1. All tables are generated directly from the public artifact by
2166 `scripts/reporting/make_v2_tables.py`.
2167

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2169

2170 **Declarations**

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2172

2173 **Availability of data and materials.** This study is based on the public EMBER
2174 Anderson and Roth (2018), UNSW-NB15 Moustafa and Slay (2015), and ToN-IoT
2175 Alsaedi et al. (2020) benchmark datasets. The datasets supporting the conclusions
2176 of this article—derived split definitions, shift constructions, experiment manifests,
2177 aggregated results, and summary tables—are available in the Zenodo repository,
2178 <https://doi.org/10.5281/zenodo.19147649>. All code used to generate the splits, run
2179 the classical and quantum kernel experiments, and reproduce every table and figure
2180 in this article is part of the same artifact:
2181

2182

2183 • Project name: Controlled Kernel Evaluation under Distribution Shift

2184

2185 • Project home page: [https://github.com/roberto-fernandez-barrios/kernel_shift_](https://github.com/roberto-fernandez-barrios/kernel_shift_framework)
2186 [framework](https://github.com/roberto-fernandez-barrios/kernel_shift_framework)

2187

2188 • Archived version: <https://doi.org/10.5281/zenodo.19147649>

2189

2190 • Operating system(s): platform independent

2191

2192 • Programming language: Python (3.12)

2193

2194 • Other requirements: pinned environment provided (`environment.lock.yml`); qiskit
2195 2.3.1

2196

2197 • License: BSD-3-Clause

2198

2199 • Any restrictions to use by non-academics: none

2200

2201 **Competing interests.** The authors declare that they have no competing interests.

2202

2203 **Funding.** The authors received no specific funding for the submitted work.

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2205

Table A1 EMBER, SVC: full Best-by-ODD comparison against the extended classical family.

Shift	Seed	Size	Classical	Quantum	OOD C	OOD Q	Δ_{OOD}	Eff.
m1	42	S	Laplacian (d4)	PauliXZ (d12)	0.756 ± 0.007	0.758 ± 0.020	+0.002	+0.09
m1	42	M	Laplacian (d12)	ZZ-r1 (d12)	0.753 ± 0.019	0.761 ± 0.007	+0.009	+0.42
m1	42	L	Laplacian (d12)	ZZ-r1 (d12)	0.750 ± 0.014	0.757 ± 0.008	+0.006	+0.38
m1	123	S	Laplacian (d12)	PauliXZ (d12)	0.762 ± 0.019	0.750 ± 0.020	-0.012	-0.43
m1	123	M	Laplacian (d12)	ZZ-r1 (d10)	0.760 ± 0.027	0.756 ± 0.029	-0.003	-0.08
m1	123	L	Laplacian (d12)	ZZ-r1 (d10)	0.751 ± 0.019	0.753 ± 0.014	+0.002	+0.09
m1	999	S	Laplacian (d12)	PauliXZ (d12)	0.767 ± 0.028	0.768 ± 0.022	+0.000	+0.00
m1	999	M	Matérn-5/2 (d12)	ZZ-r1 (d12)	0.755 ± 0.015	0.762 ± 0.013	+0.007	+0.34
m1	999	L	RBF (d12)	ZZ-r1 (d10)	0.763 ± 0.024	0.753 ± 0.005	-0.010	-0.42
m2	42	S	Laplacian (d10)	ZZ-r1 (d10)	0.762 ± 0.012	0.763 ± 0.018	+0.000	+0.02
m2	42	M	Laplacian (d12)	ZZ-r2 (d12)	0.767 ± 0.012	0.784 ± 0.009	+0.017	+1.10
m2	42	L	Laplacian (d12)	ZZ-r2 (d12)	0.783 ± 0.008	0.795 ± 0.011	+0.012	+0.85
m2	123	S	Laplacian (d12)	ZZ-r1 (d12)	0.762 ± 0.023	0.771 ± 0.019	+0.009	+0.30
m2	123	M	Laplacian (d12)	ZZ-r1 (d12)	0.775 ± 0.017	0.783 ± 0.013	+0.007	+0.35
m2	123	L	Laplacian (d12)	ZZ-r2 (d12)	0.785 ± 0.011	0.794 ± 0.007	+0.009	+0.71
m2	999	S	Laplacian (d12)	ZZ-r1 (d12)	0.769 ± 0.014	0.775 ± 0.022	+0.006	+0.25
m2	999	M	Laplacian (d10)	ZZ-r2 (d12)	0.776 ± 0.016	0.780 ± 0.020	+0.004	+0.14
m2	999	L	Laplacian (d12)	ZZ-r2 (d12)	0.787 ± 0.013	0.803 ± 0.009	+0.015	+0.97

Table A2 EMBER, GP classifier: full Best-by-ODD comparison against the extended classical family.

Shift	Seed	Size	Classical	Quantum	OOD C	OOD Q	Δ_{OOD}	Eff.
m1	42	S	Laplacian (d12)	PauliXZ (d12)	0.767 ± 0.018	0.760 ± 0.017	-0.007	-0.29
m1	42	M	Laplacian (d12)	ZZ-r1 (d12)	0.765 ± 0.021	0.766 ± 0.011	+0.002	+0.07
m1	42	L	RBF (d10)	PauliXZ (d12)	0.764 ± 0.015	0.761 ± 0.011	-0.003	-0.16
m1	123	S	Laplacian (d12)	PauliXZ (d12)	0.765 ± 0.016	0.753 ± 0.025	-0.012	-0.41
m1	123	M	Laplacian (d12)	PauliXZ (d12)	0.775 ± 0.035	0.772 ± 0.023	-0.004	-0.09
m1	123	L	Matérn-5/2 (d10)	ZZ-r2 (d12)	0.758 ± 0.017	0.754 ± 0.014	-0.004	-0.18
m1	999	S	Laplacian (d12)	PauliXZ (d12)	0.766 ± 0.028	0.762 ± 0.027	-0.004	-0.11
m1	999	M	Matérn-5/2 (d12)	Z-map (d12)	0.768 ± 0.016	0.776 ± 0.009	+0.008	+0.43
m1	999	L	RBF (d12)	ZZ-r1 (d8)	0.763 ± 0.009	0.754 ± 0.008	-0.008	-0.74
m2	42	S	Laplacian (d12)	ZZ-r1 (d12)	0.762 ± 0.024	0.764 ± 0.015	+0.002	+0.06
m2	42	M	Laplacian (d12)	ZZ-r2 (d12)	0.767 ± 0.013	0.778 ± 0.008	+0.010	+0.68
m2	42	L	Laplacian (d12)	ZZ-r2 (d12)	0.783 ± 0.008	0.793 ± 0.009	+0.010	+0.85
m2	123	S	Laplacian (d12)	ZZ-r2 (d12)	0.767 ± 0.025	0.772 ± 0.015	+0.005	+0.19
m2	123	M	Laplacian (d12)	ZZ-r1 (d12)	0.772 ± 0.020	0.783 ± 0.015	+0.011	+0.44
m2	123	L	Laplacian (d12)	ZZ-r2 (d12)	0.780 ± 0.010	0.791 ± 0.010	+0.012	+0.83
m2	999	S	Laplacian (d12)	ZZ-r1 (d10)	0.767 ± 0.021	0.775 ± 0.021	+0.008	+0.27
m2	999	M	Laplacian (d12)	ZZ-r2 (d12)	0.776 ± 0.016	0.780 ± 0.025	+0.004	+0.13
m2	999	L	Laplacian (d12)	ZZ-r2 (d12)	0.786 ± 0.013	0.798 ± 0.011	+0.012	+0.70

2255

2256 **Table A3** Network flows, SVC: full Best-by-OOD comparison against the extended classical family.

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2258	Shift	Seed	Size	Classical	Quantum	OOD C	OOD Q	Δ_{OOD}	Eff.
2259	TS/m2	42	S	Laplacian (d12)	ZZ-r1 (d6)	0.941 ± 0.013	0.954 ± 0.009	+0.014	+0.90
2260	TS/m2	42	M	Laplacian (d12)	ZZ-r1 (d10)	0.941 ± 0.014	0.958 ± 0.005	+0.017	+1.15
2261	TS/m2	42	L	Laplacian (d12)	ZZ-r1 (d8)	0.959 ± 0.009	0.964 ± 0.006	+0.005	+0.44
2262	TS/m2	123	S	Laplacian (d12)	ZZ-r1 (d8)	0.943 ± 0.017	0.959 ± 0.011	+0.015	+0.75
2263	TS/m2	123	M	Laplacian (d12)	ZZ-r1 (d6)	0.954 ± 0.007	0.958 ± 0.006	+0.003	+0.35
2264	TS/m2	123	L	Laplacian (d10)	ZZ-r1 (d8)	0.955 ± 0.003	0.959 ± 0.003	+0.003	+0.72
2265	TS/m2	999	S	Laplacian (d10)	ZZ-r1 (d10)	0.936 ± 0.011	0.957 ± 0.003	+0.021	+1.80
2266	TS/m2	999	M	Laplacian (d12)	ZZ-r1 (d8)	0.954 ± 0.009	0.962 ± 0.007	+0.008	+0.72
2267	TS/m2	999	L	Laplacian (d12)	ZZ-r1 (d10)	0.965 ± 0.004	0.969 ± 0.003	+0.003	+0.71
2268	TS/nat	42	S	Laplacian (d8)	ZZ-r1 (d10)	0.987 ± 0.003	0.988 ± 0.005	+0.001	+0.10
2269	TS/nat	42	M	Laplacian (d10)	ZZ-r1 (d12)	0.989 ± 0.004	0.990 ± 0.002	+0.001	+0.24
2270	TS/nat	42	L	Laplacian (d12)	Z-map (d12)	0.989 ± 0.005	0.991 ± 0.003	+0.002	+0.31
2271	TS/nat	123	S	Laplacian (d8)	ZZ-r1 (d12)	0.990 ± 0.005	0.992 ± 0.005	+0.002	+0.36
2272	TS/nat	123	M	Laplacian (d12)	Z-map (d10)	0.989 ± 0.003	0.992 ± 0.002	+0.003	+0.61
2273	TS/nat	123	L	Laplacian (d10)	Z-map (d12)	0.990 ± 0.002	0.991 ± 0.003	+0.002	+0.40
2274	TS/nat	999	S	Laplacian (d12)	Z-map (d12)	0.979 ± 0.006	0.985 ± 0.002	+0.006	+1.05
2275	TS/nat	999	M	Laplacian (d10)	Z-map (d10)	0.990 ± 0.002	0.991 ± 0.004	+0.002	+0.38
2276	TS/nat	999	L	Laplacian (d10)	Z-map (d12)	0.992 ± 0.002	0.992 ± 0.003	+0.000	+0.12
2277	UD/m2	42	S	Laplacian (d12)	PauliXZ (d12)	0.920 ± 0.016	0.919 ± 0.014	-0.001	-0.06
2278	UD/m2	42	M	Laplacian (d12)	PauliXZ (d12)	0.934 ± 0.017	0.937 ± 0.007	+0.003	+0.16
2279	UD/m2	42	L	Laplacian (d12)	PauliXZ (d10)	0.942 ± 0.008	0.940 ± 0.006	-0.002	-0.18
2280	UD/m2	123	S	Laplacian (d12)	PauliXZ (d12)	0.917 ± 0.020	0.926 ± 0.013	+0.008	+0.33
2281	UD/m2	123	M	Laplacian (d12)	PauliXZ (d12)	0.943 ± 0.007	0.940 ± 0.010	-0.004	-0.28
2282	UD/m2	123	L	Laplacian (d12)	Z-map (d12)	0.948 ± 0.003	0.947 ± 0.005	-0.001	-0.20
2283	UD/m2	999	S	Laplacian (d12)	PauliXZ (d12)	0.923 ± 0.023	0.927 ± 0.009	+0.004	+0.16
2284	UD/m2	999	M	Laplacian (d12)	PauliXZ (d12)	0.935 ± 0.013	0.939 ± 0.007	+0.004	+0.26
2285	UD/m2	999	L	Laplacian (d12)	Z-map (d12)	0.947 ± 0.004	0.946 ± 0.004	-0.001	-0.22
2286	UD/nat	42	S	Linear (d4)	ZZ-r2 (d10)	0.781 ± 0.040	0.809 ± 0.017	+0.028	+0.65
2287	UD/nat	42	M	Laplacian (d12)	Z-map (d12)	0.829 ± 0.019	0.838 ± 0.014	+0.009	+0.38
2288	UD/nat	42	L	Laplacian (d12)	Z-map (d12)	0.829 ± 0.011	0.829 ± 0.008	+0.000	+0.01
2289	UD/nat	123	S	Laplacian (d12)	ZZ-r2 (d6)	0.775 ± 0.026	0.818 ± 0.018	+0.043	+1.37
2290	UD/nat	123	M	Laplacian (d12)	ZZ-r1 (d10)	0.790 ± 0.020	0.812 ± 0.007	+0.022	+1.04
2291	UD/nat	123	L	Laplacian (d12)	ZZ-r1 (d12)	0.809 ± 0.012	0.825 ± 0.011	+0.016	+0.97
2292	UD/nat	999	S	Linear (d4)	ZZ-r2 (d6)	0.759 ± 0.051	0.800 ± 0.018	+0.042	+0.77
2293	UD/nat	999	M	Laplacian (d12)	ZZ-r2 (d10)	0.788 ± 0.016	0.809 ± 0.014	+0.021	+0.95
2294	UD/nat	999	L	Laplacian (d12)	ZZ-r1 (d12)	0.804 ± 0.006	0.826 ± 0.013	+0.023	+1.63
2295	UR/m2	42	S	Laplacian (d12)	Z-map (d12)	0.909 ± 0.025	0.909 ± 0.015	-0.000	-0.01
2296	UR/m2	42	M	Laplacian (d12)	PauliXZ (d12)	0.931 ± 0.019	0.935 ± 0.008	+0.004	+0.18
2297	UR/m2	42	L	Laplacian (d12)	PauliXZ (d12)	0.944 ± 0.005	0.943 ± 0.006	-0.001	-0.12
2298	UR/m2	123	S	Laplacian (d12)	PauliXZ (d12)	0.946 ± 0.014	0.942 ± 0.007	-0.004	-0.24
2299	UR/m2	123	M	Laplacian (d12)	PauliXZ (d12)	0.951 ± 0.011	0.948 ± 0.008	-0.004	-0.26
2300	UR/m2	123	L	Laplacian (d12)	Z-map (d12)	0.954 ± 0.007	0.954 ± 0.006	-0.000	-0.03
	UR/m2	999	S	Laplacian (d12)	Z-map (d12)	0.915 ± 0.016	0.918 ± 0.014	+0.003	+0.12
	UR/m2	999	M	Laplacian (d12)	Z-map (d12)	0.947 ± 0.006	0.944 ± 0.008	-0.003	-0.33
	UR/m2	999	L	Laplacian (d12)	Z-map (d12)	0.956 ± 0.003	0.948 ± 0.007	-0.008	-1.03
	UR/nat	42	S	Linear (d12)	ZZ-r2 (d6)	0.770 ± 0.019	0.778 ± 0.013	+0.008	+0.36
	UR/nat	42	M	Laplacian (d10)	ZZ-r1 (d4)	0.809 ± 0.035	0.815 ± 0.010	+0.007	+0.18
	UR/nat	42	L	Laplacian (d10)	ZZ-r1 (d4)	0.812 ± 0.020	0.815 ± 0.008	+0.003	+0.12
	UR/nat	123	S	Linear (d12)	ZZ-r1 (d6)	0.795 ± 0.012	0.797 ± 0.018	+0.002	+0.11
	UR/nat	123	M	Linear (d10)	ZZ-r1 (d4)	0.800 ± 0.018	0.802 ± 0.024	+0.003	+0.09
	UR/nat	123	L	Laplacian (d12)	ZZ-r1 (d4)	0.818 ± 0.018	0.811 ± 0.009	-0.007	-0.34
	UR/nat	999	S	Linear (d12)	ZZ-r1 (d6)	0.780 ± 0.020	0.781 ± 0.008	+0.000	+0.02
	UR/nat	999	M	Linear (d12)	ZZ-r2 (d6)	0.783 ± 0.014	0.809 ± 0.015	+0.026	+1.26
	UR/nat	999	L	Laplacian (d8)	ZZ-r2 (d6)	0.804 ± 0.015	0.814 ± 0.008	+0.010	+0.59

Table A4 Network flows, GP classifier: full Best-by-OOD comparison against the extended classical family.

Shift	Seed	Size	Classical	Quantum	OOD C	OOD Q	Δ_{OOD}	Eff.
TS/m2	42	S	Laplacian (d12)	ZZ-r1 (d8)	0.906 ± 0.019	0.954 ± 0.011	+0.049	+2.27
TS/m2	42	M	Laplacian (d8)	ZZ-r1 (d12)	0.921 ± 0.004	0.957 ± 0.005	+0.037	+5.70
TS/m2	42	L	Laplacian (d8)	ZZ-r1 (d8)	0.931 ± 0.007	0.964 ± 0.006	+0.034	+3.58
TS/m2	123	S	Laplacian (d8)	ZZ-r1 (d10)	0.910 ± 0.056	0.958 ± 0.006	+0.048	+0.86
TS/m2	123	M	Laplacian (d4)	ZZ-r1 (d12)	0.932 ± 0.007	0.958 ± 0.007	+0.025	+2.59
TS/m2	123	L	Laplacian (d4)	ZZ-r1 (d10)	0.934 ± 0.004	0.960 ± 0.003	+0.026	+4.84
TS/m2	999	S	Laplacian (d6)	ZZ-r1 (d10)	0.918 ± 0.021	0.954 ± 0.009	+0.036	+1.58
TS/m2	999	M	Laplacian (d4)	ZZ-r1 (d10)	0.931 ± 0.002	0.961 ± 0.006	+0.030	+4.49
TS/m2	999	L	Laplacian (d8)	ZZ-r1 (d10)	0.941 ± 0.003	0.968 ± 0.003	+0.027	+6.65
TS/nat	42	S	Laplacian (d8)	ZZ-r1 (d8)	0.981 ± 0.005	0.990 ± 0.002	+0.009	+1.67
TS/nat	42	M	Laplacian (d8)	ZZ-r1 (d10)	0.987 ± 0.002	0.991 ± 0.003	+0.004	+1.08
TS/nat	42	L	Laplacian (d8)	ZZ-r1 (d12)	0.983 ± 0.003	0.992 ± 0.002	+0.009	+2.30
TS/nat	123	S	Laplacian (d8)	ZZ-r1 (d8)	0.981 ± 0.017	0.992 ± 0.003	+0.011	+0.65
TS/nat	123	M	Laplacian (d8)	ZZ-r1 (d10)	0.985 ± 0.004	0.993 ± 0.002	+0.008	+1.72
TS/nat	123	L	Laplacian (d8)	ZZ-r1 (d12)	0.985 ± 0.004	0.992 ± 0.002	+0.007	+1.64
TS/nat	999	S	Laplacian (d10)	ZZ-r1 (d10)	0.965 ± 0.027	0.980 ± 0.015	+0.015	+0.48
TS/nat	999	M	Laplacian (d8)	ZZ-r1 (d10)	0.984 ± 0.005	0.991 ± 0.003	+0.007	+1.20
TS/nat	999	L	Laplacian (d8)	ZZ-r1 (d10)	0.988 ± 0.002	0.992 ± 0.003	+0.004	+1.41
UD/m2	42	S	Laplacian (d12)	PauliXZ (d12)	0.894 ± 0.028	0.904 ± 0.016	+0.009	+0.29
UD/m2	42	M	Laplacian (d12)	PauliXZ (d12)	0.911 ± 0.007	0.934 ± 0.009	+0.023	+1.96
UD/m2	42	L	Laplacian (d12)	Z-map (d12)	0.927 ± 0.008	0.939 ± 0.010	+0.013	+1.02
UD/m2	123	S	Laplacian (d12)	Z-map (d12)	0.889 ± 0.019	0.917 ± 0.019	+0.028	+1.04
UD/m2	123	M	Laplacian (d12)	PauliXZ (d12)	0.916 ± 0.016	0.937 ± 0.007	+0.022	+1.24
UD/m2	123	L	Laplacian (d12)	Z-map (d12)	0.932 ± 0.014	0.945 ± 0.007	+0.013	+0.86
UD/m2	999	S	Laplacian (d12)	Z-map (d12)	0.878 ± 0.022	0.911 ± 0.031	+0.033	+0.87
UD/m2	999	M	Laplacian (d12)	PauliXZ (d12)	0.914 ± 0.011	0.931 ± 0.009	+0.017	+1.19
UD/m2	999	L	Laplacian (d12)	PauliXZ (d12)	0.936 ± 0.009	0.944 ± 0.003	+0.008	+0.82
UD/nat	42	S	Laplacian (d12)	ZZ-r2 (d6)	0.764 ± 0.023	0.801 ± 0.023	+0.037	+1.13
UD/nat	42	M	Laplacian (d12)	PauliXZ (d12)	0.813 ± 0.010	0.829 ± 0.008	+0.016	+1.20
UD/nat	42	L	Laplacian (d12)	Z-map (d12)	0.818 ± 0.010	0.831 ± 0.009	+0.013	+0.97
UD/nat	123	S	Laplacian (d12)	ZZ-r2 (d10)	0.775 ± 0.027	0.813 ± 0.017	+0.038	+1.20
UD/nat	123	M	Laplacian (d12)	ZZ-r2 (d6)	0.781 ± 0.020	0.805 ± 0.016	+0.024	+0.93
UD/nat	123	L	Laplacian (d12)	ZZ-r1 (d12)	0.800 ± 0.005	0.816 ± 0.006	+0.016	+2.09
UD/nat	999	S	Linear (d4)	ZZ-r2 (d6)	0.766 ± 0.053	0.797 ± 0.021	+0.032	+0.55
UD/nat	999	M	Laplacian (d12)	ZZ-r2 (d12)	0.784 ± 0.018	0.806 ± 0.013	+0.022	+1.02
UD/nat	999	L	Laplacian (d12)	ZZ-r2 (d12)	0.805 ± 0.006	0.822 ± 0.012	+0.017	+1.31
UR/m2	42	S	Laplacian (d12)	Z-map (d12)	0.880 ± 0.021	0.902 ± 0.022	+0.022	+0.73
UR/m2	42	M	Laplacian (d12)	Z-map (d12)	0.901 ± 0.008	0.925 ± 0.009	+0.024	+1.99
UR/m2	42	L	Laplacian (d12)	PauliXZ (d12)	0.929 ± 0.012	0.942 ± 0.003	+0.013	+1.05
UR/m2	123	S	Laplacian (d12)	Z-map (d12)	0.905 ± 0.012	0.926 ± 0.023	+0.021	+0.81
UR/m2	123	M	Laplacian (d12)	Z-map (d12)	0.914 ± 0.010	0.946 ± 0.010	+0.032	+2.29
UR/m2	123	L	Laplacian (d12)	PauliXZ (d12)	0.946 ± 0.004	0.953 ± 0.006	+0.007	+0.91
UR/m2	999	S	Laplacian (d10)	PauliXZ (d12)	0.883 ± 0.023	0.910 ± 0.025	+0.027	+0.81
UR/m2	999	M	Laplacian (d12)	Z-map (d12)	0.912 ± 0.004	0.941 ± 0.008	+0.029	+3.13
UR/m2	999	L	Laplacian (d10)	PauliXZ (d12)	0.937 ± 0.014	0.949 ± 0.003	+0.012	+0.84
UR/nat	42	S	Poly-2 (d4)	PauliXZ (d6)	0.760 ± 0.020	0.784 ± 0.024	+0.025	+0.79
UR/nat	42	M	Laplacian (d10)	ZZ-r1 (d4)	0.795 ± 0.040	0.817 ± 0.013	+0.022	+0.51
UR/nat	42	L	Laplacian (d8)	ZZ-r1 (d4)	0.796 ± 0.007	0.813 ± 0.005	+0.017	+1.95
UR/nat	123	S	Poly-2 (d12)	ZZ-r2 (d6)	0.780 ± 0.024	0.797 ± 0.016	+0.017	+0.59
UR/nat	123	M	Linear (d12)	ZZ-r1 (d4)	0.794 ± 0.014	0.803 ± 0.020	+0.008	+0.34
UR/nat	123	L	Laplacian (d12)	ZZ-r2 (d12)	0.810 ± 0.023	0.812 ± 0.019	+0.002	+0.06
UR/nat	999	S	Poly-2 (d12)	ZZ-r2 (d6)	0.777 ± 0.016	0.782 ± 0.005	+0.005	+0.30
UR/nat	999	M	Poly-2 (d12)	ZZ-r2 (d6)	0.780 ± 0.017	0.814 ± 0.016	+0.034	+1.51
UR/nat	999	L	Laplacian (d8)	ZZ-r1 (d4)	0.790 ± 0.010	0.807 ± 0.011	+0.017	+1.10

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 2348 curation, formal analysis, visualization, investigation, writing—original draft,
 2349 writing—review and editing. IPL: supervision, validation, writing—review and editing.
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